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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A140NHT11GQ0012-HP161H2 Stroage 1st stage KO drum7 Percent.pvdb

Date of Analysis : Sep 17,2026 2:03pm

PV Elite 28 U1, March 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 900.000 mm. ID	
Vessel Design Length, Tangent to Tangent	3000.00 mm.
Distance of Bottom Tangent above Grade	3100.00 mm.
Distance of Base above Grade	300.00 mm.
Specified Datum Line Distance	2850.00 mm.
Internal Design Temperature	120 °C
Internal Design Pressure	32.500 bars
External Design Temperature	120 °C
External Design Pressure	1.034 bars
Maximum Allowable Working Pressure	37 bars
External Max. Allowable Working Pressure	15 bars
Shop Test Pressure	47.655 bars
Field Test Pressure	47.655 bars
Required Minimum Design Metal Temperature	0.0 °C
Warmest Computed Minimum Design Metal Temperature	-20.0 °C
Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N
Head	SA-516 70	...	...	K02700	+N
Skirt	SA-516 70	...	...	K02700	+N
Nozzle	SA-105	...	...	K03504	
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-283 C	...	...	K02401	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt 1	...	...	...	1.5000	N/A	No
Skirt 2	...	...	...	1.5000	N/A	No
Bottom D end	32.594	1.03	37.000	2.0000	No	No
Shell	32.568	1.03	54.000	2.0000	N/A	No
Top D end	32.500	1.03	49.000	2.0000	No	No

Liquid Level: 975.00 mm. Dens.: 0.001 kg./cm3 Sp. Gr.: 0.986

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Skirt	-1050.0	1800.0	8.0	8.0	...	...	0.70	0.55
Skirt	-50.0	1000.0	8.0	8.0	...	...	0.70	0.55
Ellipse	-0.0	50.0	16.0	14.0	12.6	4.4	1.00	1.00

Cylinder	2900.0	2900.0	20.0	20.0	12.8	7.0	1.00	1.00
Ellipse	2950.0	50.0	22.0	18.0	12.6	4.4	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1_16"	WRC-297	0.831	Shell	Passed
N5_8"	WRC-297	0.507	Shell	Passed
N2_16"	WRC-107/537	0.674	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	900	3317.38	1840.95	16970.5	9067.71
20	30	2300	3055.05	1816.19	11235.2	5776.22
30	40	2825	2909.32	1770.06	8252.91	3983.07
40	50	4300	2902.03	1729.91	8107.63	3895.56
50	60	5775	2473.48	879.85	312.982	111.332

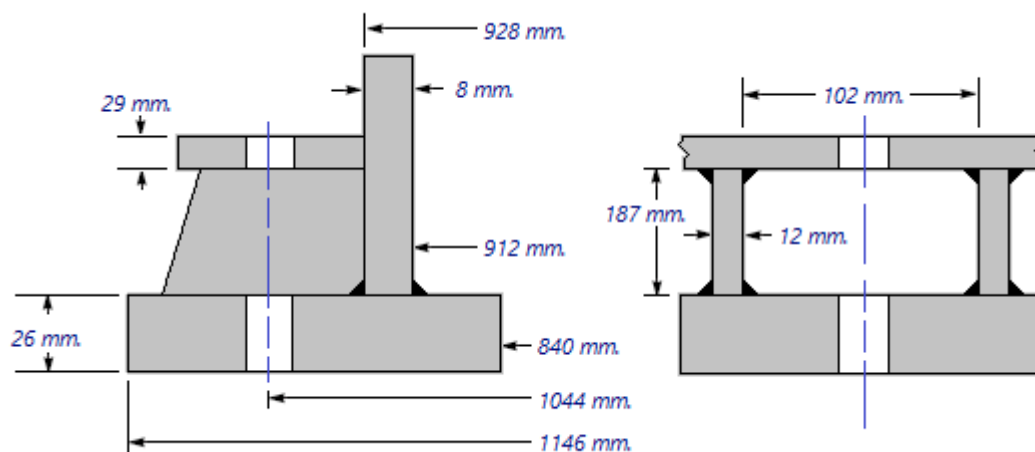
Abs Max of the all of the Stress Ratio's : 0.4289

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing 26.0000 mm.
 Inside Diameter of Basing 840.0000 mm.
 Outside Diameter of Basing 1146.0000 mm.
 Nominal Diameter of Bolts 30.0000 mm.
 Diameter of Bolt Circle 1044.0000 mm.
 Number of Bolts 8

Thickness of Gusset Plates 12.0000 mm.
 Average Width of Gusset Plates 109.0000 mm.
 Height of Gussets 187.0000 mm.
 Distance between Gussets 102.0000 mm.
 Thickness of Top Plate or Ring 29.0000 mm.
 Radial Width of the Top Plate 109.0000 mm.
 Number of Gussets 16

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:



Total Wind Shear on Support	3317. Kgf
Total Earthquake Shear on Support	1841. Kgf
Wind Moment on Support	17200. Kg-m.
Earthquake Moment on Support	9298. Kg-m.
Maximum Initial Bolt Load	7249. Kgf

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	5077.1 kg.
Shop Test - Fabricated + water (full)	7330.5 kg.
Shipping - Fab. + rem. intls.+ shipping app.	7751.3 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	7906.3 kg.
Empty - Fab. + intls. + details + wghts.	7906.3 kg.
Operating - Empty + operating liquid (no ca)	8470.6 kg.
Field Test - Empty weight + test medium (full)	10004.8 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3_2"	37	...	...	OK	6.80	OK	No Calc[*]
N1_16"	38	OK	...	OK	10.33	OK	Passed
N5_8"	46	OK	...	OK	10.10	OK	Passed
K1A_3"	54	...	...	OK	6.80	OK	No Calc[*]
K1B_3"	47	OK	...	OK	7.26	OK	Passed
K2A_2"	54	...	...	OK	6.80	OK	No Calc[*]
K2B_2"	54	...	...	OK	6.80	OK	No Calc[*]
M1_24"	38	OK	...	...		OK	Passed
N2_16"	38	OK	...	OK	10.33	OK	Passed
N4_2"	49	OK	...	OK	6.80	OK	Passed
N4_2"	49	...	...	OK	6.80	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 37 Nozzle : N32"
 Minimum MAWP Shells/Flanges : 37 Element : Bottom D end
 Minimum MAPnc Shells/Flanges : 43 Element : Bottom D end

Computed Vessel M.A.W.P. : 37 bars

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	N3_2"	0.000	0.000	109.600
40	N1_16"	1500.000	0.000	820.800
40	N5_8"	2400.000	0.000	414.400
40	K1A_3"	800.000	0.000	120.328
40	K1B_3"	800.000	90.000	160.400
40	K2A_2"	200.000	0.000	120.328
40	K2B_2"	200.000	90.000	120.328
40	M1_24"	850.000	180.000	1227.200
50	N2_16"	0.000	0.000	820.800
50	N4_2"	0.000	0.000	116.328

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

*Diameter Limit clash between : N4_2" and N216".
 The Centerline Dist. is 418.17 while the sum of the limit Radii is 468.56 mm..*

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N3_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	215.08	300
K1A_3"	2.000 in	Actual	LWN	3.320	16.764	...	...	228.35	300
K2A_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.98	300
K2B_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.98	300
N4_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	245.58	300
K1B_3"	3.000 in	Actual	LWN	4.320	16.764	...	...	223.36	300
N5_8"	8.000 in	Actual	WNF	8.945	12.000	...	...	242.74	300
N1_16"	16.000 in	Actual	WNF	16.945	12.000	...	...	341.38	300
N2_16"	16.000 in	Actual	WNF	16.945	12.000	...	...	520.97	300
M1_24"	24.000 in	Actual	WNF	24.945	12.000	...	...	339.90	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N3_2"	SA-105	14.000	10.000	...	...	...
K1A_3"	SA-105	20.000	10.000	...	...	...
K2A_2"	SA-105	20.000	20.000	...	...	...
K2B_2"	SA-105	20.000	20.000	...	...	...
N4_2"	SA-105	18.000	10.000	...	...	...
K1B_3"	SA-105	20.000	10.000	...	...	...
N5_8"	SA-105	20.000	10.000	...	...	...
N1_16"	SA-105	20.000	10.000	...	...	...
N2_16"	SA-105	18.000	10.000	...	...	...
M1_24"	SA-105	14.000	14.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
N3_2"	...	0.0	200.00	0.00	Bottom D end
K1A_3"	750.000	0.0	206.38	0.00	Shell
K2A_2"	150.000	0.0	200.00	0.00	Shell
K2B_2"	150.000	90.0	200.00	0.00	Shell
N4_2"	...	0.0	200.00	0.00	Top D end
K1B_3"	750.000	90.0	200.00	0.00	Shell
N5_8"	2350.000	0.0	200.00	0.00	Shell
N1_16"	1450.000	0.0	250.00	0.00	Shell
N2_16"	...	0.0	462.47	0.00	Top D end
M1_24"	800.000	180.0	133.38	0.00	Shell



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SKIRT: 8.0mm. THK X 912.0mm. ID X 1800.0mm.	SA-516 70
1	SKIRT: 8.0mm. THK X 912.0mm. ID X 1000.0mm.	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 16.0mm. THK X 900.0mm. ID X 50.0mm.	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 22.0mm. THK X 900.0mm. ID X 50.0mm.	SA-516 70
1	PLATFORM: 1000mm X 3600mm WIDE	
1	LADDER: 5900.0mm	...
2	CLASS 300 GR 1.1, 2.0" BLIND FLANGE(S)	SA-105
1	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 1146.0mm. OD X 840.0 ID X 26.0mm.	SA-283 C
8	BASERING BOLTS: 30.0mm. DIA	SA-36
8	BASERING BOLT NUTS: 30.0mm. DIA	...
1	BASERING TOP RING: 1146.0mm. OD X 928.0 ID X 29.0mm.	...
16	BASERING GUSSET PLATES: 109.0mm. X 109.0mm. X 187.0mm. X 12.0mm.	...
1	CYLINDRICAL SEGMENT 20.0mm. THK X 1570.8mm. X 2900.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm. THK X 1256.6mm. X 2900.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Notes								
Bottom D end [10]	B	-35	-35		1.000	14.000	1.00	No
Bottom D end [7]	B	-31	-38		0.874	16.000	1.00	No
Shell [8]	B	-24	-42	-29	0.679	20.000	1.00	No
Top D end [10]	B	-27	-41	-29	0.748	18.000	1.00	No
Top D end [7]	B	-21	-43	-29	0.610	22.000	1.00	No
N32"	[1]	A	-15	-20	0.901	14.000	1.00	No
Nozzle Flg [5]	A	-18	-33					
N116"	[1]	A	-4	-22	0.678	20.000	1.00	No
Nozzle Flg [5]	A	-18	-33					
N58"	[1]	A	-19	-104	0.280	12.000	1.00	No
Nozzle Flg [5]	A	-18	-33	-29				
K1A3"	[1]	A	-9	-104	0.050	16.764	1.00	No
Nozzle Flg [5]	A	-18	-33					
K1B3"	[1]	A	-9	-104	0.073	16.764	1.00	No
Nozzle Flg [5]	A	-18	-33					
K2A2"	[1]	A	-9	-104	0.050	16.764	1.00	No
Nozzle Flg [5]	A	-18	-33					
K2B2"	[1]	A	-9	-104	0.050	16.764	1.00	No
Nozzle Flg [5]	A	-18	-33					
M124"	[1]	A	-4	-22	0.678	20.000	1.00	No
Nozzle Flg [5]	A	-18	-33					
N216"	[1]	A	-7	-21	0.748	18.000	1.00	No
Nozzle Flg [5]	A	-18	-33					
N4_2"	[1]	A	-9	-104	0.050	16.764	1.00	No
Nozzle Flg [5]	A	-18	-33					
Noz1. Bolting[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: -4 -20

Required Minimum Design Metal Temperature 0.0 °C
 Warmest Computed Minimum Design Metal Temperature -20.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and



UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note	20	40	There is a high jump in the Joint Eff.
Note	10	20	The wind diameter multiplier is very big.
Note	20	30	The wind diameter multiplier is very big.
Note	30	40	The wind diameter multiplier is very big.
Note	40	50	The wind diameter multiplier is very big.
Note	50	60	The wind diameter multiplier is very big.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	32.5	bars
Design Internal Temperature	120.0	°C
Projection of Nozzle from Vessel Top	500	mm.
Projection of Nozzle from Vessel Bottom	500	mm.
Minimum Design Metal Temperature	0.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW
Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm.
Distance Upwind of Crest	Lh 0 mm.
Distance from Crest to the Vessel	x 0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.
Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	1.500
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	No
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg
M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:



Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt 1
Distance "FROM" to "TO"	1800 mm.
Skirt Outside Diameter	928 mm.
Diameter of Skirt at Base	928 mm.
Skirt Thickness	8 mm.
Internal Corrosion Allowance	0.75 mm.
Nominal Thickness	8 mm.
External Corrosion Allowance	0.75 mm.
Design Temperature Internal Pressure	120 °C
Design Temperature External Pressure	120 °C
Effective Diameter Multiplier	1.75
Material Name	SA-516 70 +N
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	138 N./mm2
Allowable Stress, Hydrotest	234 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	235.8 N./mm2
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.55
Weld is pre-Heated	No

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt 2
Distance "FROM" to "TO"	1000 mm.
Skirt Outside Diameter	928 mm.
Diameter of Skirt at Base	928 mm.
Skirt Thickness	8 mm.
Internal Corrosion Allowance	0.75 mm.
Nominal Thickness	8 mm.
External Corrosion Allowance	0.75 mm.
Design Temperature Internal Pressure	120 °C
Design Temperature External Pressure	120 °C
Effective Diameter Multiplier	1.75
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.55
Weld is pre-Heated	No

Element#	3/5
Element From Node	30
Element To Node	40
Element Type	Elliptical
Description	Bottom D end
Distance "FROM" to "TO"	50 mm.
Inside Diameter	900 mm.
Element Thickness	14 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	16 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	32.5 bars
Design Temperature Internal Pressure	120 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	120 °C
Effective Diameter Multiplier	1.75



Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid 30
Dist. from "FROM" Node / Offset dist	-225 mm.
Height/Length of Liquid	275 mm.
Liquid Density	0.0009844 kg./cm3
Element From Node	30
Detail Type	Nozzle
Detail ID	N32"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element#	4/5
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	2900 mm.
Inside Diameter	900 mm.
Element Thickness	20 mm.
Internal Corrosion Allowance	2 mm.
Nominal Thickness	20 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	32.5 bars
Design Temperature Internal Pressure	120 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	120 °C
Effective Diameter Multiplier	1.75
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	700 mm.
Liquid Density	0.0009844 kg./cm3
Element From Node	40
Detail Type	Nozzle
Detail ID	N116"
Dist. from "FROM" Node / Offset dist	1450 mm.
Nozzle Diameter	16 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	175.24 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	40



Detail Type	Nozzle	
Detail ID	N58"	
Dist. from "FROM" Node / Offset dist	2350	mm.
Nozzle Diameter	8	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	60.512	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K1A3"	
Dist. from "FROM" Node / Offset dist	750	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.7787	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K1B3"	
Dist. from "FROM" Node / Offset dist	750	mm.
Nozzle Diameter	3	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	16.504	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K2A2"	
Dist. from "FROM" Node / Offset dist	150	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K2B2"	
Dist. from "FROM" Node / Offset dist	150	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	M124"	
Dist. from "FROM" Node / Offset dist	800	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	



Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	40	
Detail Type	Weight	
Detail ID	Rung	
Dist. from "FROM" Node / Offset dist	150	mm.
Miscellaneous Weight	3	Kgf
Offset from Element Centerline	400	mm.
Element From Node	40	
Detail Type	Weight	
Detail ID	Rung	
Dist. from "FROM" Node / Offset dist	300	mm.
Miscellaneous Weight	3	Kgf
Offset from Element Centerline	400	mm.
Element From Node	40	
Detail Type	Weight	
Detail ID	Rung	
Dist. from "FROM" Node / Offset dist	1200	mm.
Miscellaneous Weight	3	Kgf
Offset from Element Centerline	400	mm.
Element From Node	40	
Detail Type	Weight	
Detail ID	Rung	
Dist. from "FROM" Node / Offset dist	1450	mm.
Miscellaneous Weight	3	Kgf
Offset from Element Centerline	400	mm.
Element From Node	40	
Detail Type	Weight	
Detail ID	Pipe Davit	
Dist. from "FROM" Node / Offset dist	2600	mm.
Miscellaneous Weight	400	Kgf
Offset from Element Centerline	450	mm.
Element From Node	40	
Detail Type	Weight	
Detail ID	Lifting Wt	
Dist. from "FROM" Node / Offset dist	1450	mm.
Miscellaneous Weight	150	Kgf
Offset from Element Centerline	0	mm.
Element From Node	40	
Detail Type	Weight	
Detail ID	Clip Wt	
Dist. from "FROM" Node / Offset dist	1600	mm.
Miscellaneous Weight	400	Kgf
Offset from Element Centerline	0	mm.
Element From Node	40	
Detail Type	Weight	
Detail ID	Manway davit	
Dist. from "FROM" Node / Offset dist	800	mm.
Miscellaneous Weight	50	Kgf
Offset from Element Centerline	950	mm.
Element From Node	40	
Detail Type	Weight	
Detail ID	Demister WT	
Dist. from "FROM" Node / Offset dist	2500	mm.
Miscellaneous Weight	40	Kgf
Offset from Element Centerline	0	mm.



Element From Node	40	
Detail Type	Weight	
Detail ID	N1Deflector	
Dist. from "FROM" Node / Offset dist	1450	mm.
Miscellaneous Weight	115	Kgf
Offset from Element Centerline	0	mm.

Element From Node	40	
Detail Type	Weight	
Detail ID	Demister Suppor	
Dist. from "FROM" Node / Offset dist	2500	mm.
Miscellaneous Weight	5	Kgf
Offset from Element Centerline	0	mm.

Element#	5/5	
Element From Node	50	
Element To Node	60	
Element Type	Elliptical	
Description	Top D end	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	900	mm.
Element Thickness	18	mm.
Internal Corrosion Allowance	2	mm.
Nominal Thickness	22	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	32.5	bars
Design Temperature Internal Pressure	120	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	120	°C
Effective Diameter Multiplier	1.75	
Material Name	SA-516 70 +N	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Platform	
Detail ID	PLAT 1	
Dist. from "FROM" Node / Offset dist	600	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	119955	cm2
Platform Weight	2499.3	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	3600	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	1800	mm.
Platform Force Coefficient	2.0	
Ladder Layout Angle	270.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	5900	mm.
Unit Weight of Ladder	0.05	Kgf/mm.
Platform Length (top head platform)	3600	mm.

Element From Node	50	
Detail Type	Nozzle	
Detail ID	N216"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	16	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

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XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Skirt 1	...	1800	...	1800	...
Skirt 2	...	2800	...	1000	...
Bottom D end	...	2850	...	50	...
Shell	...	5750	...	2900	...
Top D end	...	5800	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Skirt 1		...	8	1.5	928	...	1
Skirt 2		...	8	1.5	928	...	2
Bottom D end		32.594	16	2	900	138	3
Shell		32.568	20	2	900	138	4
Top D end		32.5	22	2	900	138	5

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Skirt 1		...	No Calc	No Calc	8	No Calc	1
Skirt 2		...	No Calc	No Calc	8	No Calc	2
Bottom D end		32.5	36.658	42.7969	14	12.6392	3
Shell		32.5	53.6015	59.7357	20	12.8211	4
Top D end		32.5	48.9595	54.9759	18	12.6083	5

Minimum 37 43

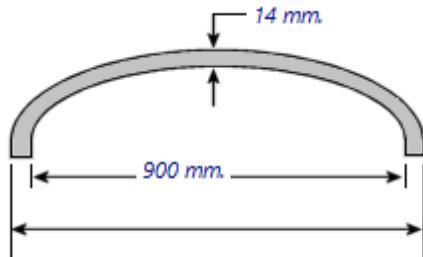
MAWP: 37 bars, limited by: Bottom D end.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom D end

Elliptical Head Design Information:					
Design Pressure	P	32.594 bars	Int. Design Temperature		120 °C
Material		SA-516 70	Ext. Design Temperature		120 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	900.000 mm.	Actual Stress		122.4 N./mm^2
Spec. Min. Thk t _e , or s _a		14.0000 mm.	Spec. Nominal Thk.		16.000 mm.
Surface Area		10879.7 cm^2	SG of Contents		0.986
Element Volume		127257.3 cm3	Weight of Contents		125.4 Kg
Empty Weight		142.4 Kg	Operating Weight		267.8 Kg
			Comp. All. Ope. Stress		135.8 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 11.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.625 mm.

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (32.594 \cdot 904.0 \cdot 0.994) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 32.594)$$

$$= 10.6392 + 2.0000 = 12.6392 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.094 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 12.0) / (0.994 \cdot 904.0 + 0.2 \cdot 12.0)$$

$$= 37 - 0.1 = 37 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 14.0) / (1.0 \cdot 900.0 + 0.2 \cdot 14.0)$$

$$= 43 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (32.594 \cdot (0.994 \cdot 904.0 + 0.2 \cdot 12.0)) / (2 \cdot 1.0 \cdot 12.0)$$

= 122.385 N./mm²

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (32.594 \cdot 452.0) / (138.0 \cdot 1.0 - 0.6 \cdot 32.594) + 2.0$$

$$= 12.830 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.072 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (452.0 + 0.6 \cdot 14.0)$$

$$= 42 - 0.1 = 42 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (904.0 / (2 \cdot 227.0)))^2 / 6$$

$$= 0.994139$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 7.453 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 14.0, tr = 12.0, c = 2.0 mm., E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 1.$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -35 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 16.0, tr = 12.234, c = 2.0 mm., E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.874$, Temp. Reduction = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -31 °C

Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	36.752 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	14.000 mm.	Basic MDMT	-35 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-35 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 11.250 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.625 mm.

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: Bottom D end

Elliptical Head From 30 to 40 Ext. Chart: CS-2 at 120 °C

Elastic Modulus from Chart: CS-2 at 120 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
12.000	928.00	77.33	0.0017960	101.27



$$MAEP = B/(K0*Do/t) = 101.2656/(0.9 * 77.3333) = 14.5488 \text{ bars}$$

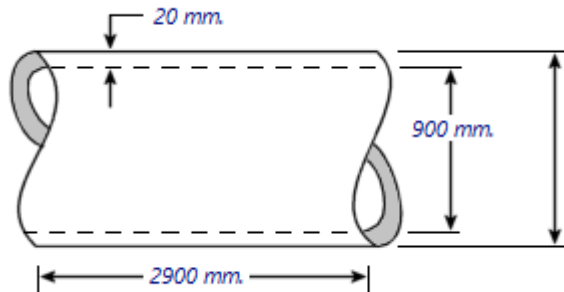
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.403	928.00	386.22	0.0003596	35.95

$$MAEP = B/(K0*Do/t) = 35.9527/(0.9 * 386.2227) = 1.0343 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:					
Design Pressure	P	32.568 bars	Int. Design Temperature		120 °C
Material		SA-516 70	Ext. Design Temperature		120 °C
Length		2900.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	900.000 mm.	Actual Stress		83.7 N./mm^2
Spec. Min. Thk t _e , or s _a		20.0000 mm.	Spec. Nominal Thk.		20.000 mm.
Surface Area		85639.8 cm^2	SG of Contents		0.986
Element Volume		1845231.0 cm3	Weight of Contents		438.9 Kg
Empty Weight		1299.3 Kg	Operating Weight		1738.1 Kg
			Comp. All. Ope. Stress		142.7 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (32.568 \cdot 452.0) / (138.0 \cdot 1.0 - 0.6 \cdot 32.568)$$

$$= 10.8211 + 2.0000 = 12.8211 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.068 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 18.0) / (452.0 + 0.6 \cdot 18.0)$$

$$= 54 - 0.1 = 54 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 20.0) / (450.0 + 0.6 \cdot 20.0)$$

$$= 60 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (32.568 \cdot (452.0 + 0.6 \cdot 18.0)) / (1.0 \cdot 18.0)$$

$$= 83.740 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.174 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 20.0, tr = 12.225, c = 2.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.679, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -24 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -42 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	36.726 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	20.000 mm.	Basic MDMT	-24 °C
Stress/Thk. Ratio	0.679	Temperature Difference	18 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-42 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 120 °C

Elastic Modulus from Chart: CS-2 at 120 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.000	940.00	3150.00	52.22	3.3511	0.0009656	84.22

$$MAEP = (4*B)/(3*(Do/t)) = (4*84.2159)/(3*52.2222) = 21.5007 \text{ bars}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.984	940.00	3150.00	188.61	3.3511	0.0001463	14.63

$$MAEP = (4*B)/(3*(Do/t)) = (4*14.6307)/(3*188.6064) = 1.0342 \text{ bars}$$

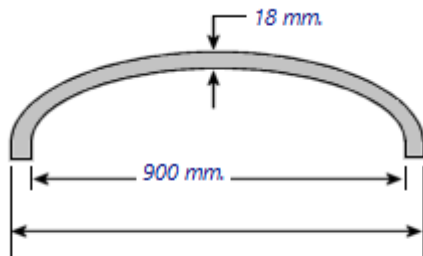
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.000	940.00	145023.56	52.22	50.0000	0.0004069	40.68

$$MAEP = (4*B)/(3*(Do/t)) = (4*40.6764)/(3*52.2222) = 10.3849 \text{ bars}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top D end

Elliptical Head Design Information:					
Design Pressure	P	32.500 bars	Int. Design Temperature		120 °C
Material		SA-516 70	Ext. Design Temperature		120 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	900.000 mm.	Actual Stress		91.6 N./mm^2
Spec. Min. Thk t _e , or sa		18.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		11142.6 cm^2	SG of Contents		0.000
Element Volume		127257.3 cm3	Weight of Contents		0.0 Kgf
Empty Weight		198.2 Kgf	Operating Weight		198.2 Kgf
			Comp. All. Ope. Stress		140.9 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 11.250 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.625 mm.

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (32.5 \cdot 904.0 \cdot 0.994) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 32.5)$$

$$= 10.6083 + 2.0000 = 12.6083 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 16.0) / (0.994 \cdot 904.0 + 0.2 \cdot 16.0)$$

$$= 49 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 18.0) / (1.0 \cdot 900.0 + 0.2 \cdot 18.0)$$

$$= 55 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (32.5 \cdot (0.994 \cdot 904.0 + 0.2 \cdot 16.0)) / (2 \cdot 1.0 \cdot 16.0)$$

$$= 91.605 \text{ N./mm}^2$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (32.5 \cdot 452.0) / (138.0 \cdot 1.0 - 0.6 \cdot 32.5) + 2.0$
 $= 12.798 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138.0 \cdot 1.0 \cdot 20.0) / (452.0 + 0.6 \cdot 20.0)$
 $= 59 \text{ bars}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$
 $= (2 + (904.0 / (2 \cdot 227.0)))^2 / 6$
 $= 0.994139$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 10.061 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 18.0$, $t_r = 11.969$, $c = 2.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.748$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-41 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 22.0$, $t_r = 12.202$, $c = 2.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.61$, Temp. Reduction = 22 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	36.658 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	18.000 mm.	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.748	Temperature Difference	14 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-41 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 11.250 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.625 mm.

Material treatment options: +N

EXTERNAL PRESSURE RESULTS for: Top D end

Elliptical Head From 50 to 60 Ext. Chart: CS-2 at 120 °C

Elastic Modulus from Chart: CS-2 at 120 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B



16.000	936.00	58.50	0.0023742	107.44
--------	--------	-------	-----------	--------

$$MAEP = B/(K0*Do/t) = 107.4417/(0.9 * 58.5) = 20.4056 \text{ bars}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.423	936.00	386.22	0.0003596	35.95

$$MAEP = B/(K0*Do/t) = 35.9531/(0.9 * 386.2186) = 1.0343 \text{ bars}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	47.655 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	42.250 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	55.450 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	40.324 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	46.475 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	46.475 bars
App 27-4	= M.A.W.P.	36.658 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 36.658 • 1.0
 = 47.655 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom D end	154.1	234.0	0.659	47.79
Shell	110.4	234.0	0.472	47.79
Top D end	120.0	234.0	0.513	47.79

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_2"	Nozzle	138.00	138.00	1.000
N1_16"	Nozzle	138.00	138.00	1.000
N5_8"	Nozzle	138.00	138.00	1.000
K1A_3"	Nozzle	138.00	138.00	1.000
K1B_3"	Nozzle	138.00	138.00	1.000
K2A_2"	Nozzle	138.00	138.00	1.000
K2B_2"	Nozzle	138.00	138.00	1.000
M1_24"	Nozzle	138.00	138.00	1.000
N2_16"	Nozzle	138.00	138.00	1.000
N4_2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom D end	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Top D end	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3_2"	10.11	225.00	0.045
N1_16"	83.80	225.00	0.372
N5_8"	43.33	225.00	0.193
K1A_3"	10.11	225.00	0.045
K1B_3"	13.73	225.00	0.061
K2A_2"	10.11	225.00	0.045
K2B_2"	10.11	225.00	0.045
M1_24"	124.27	225.00	0.552



N2_16"	83.80	225.00	0.372
N4__2"	10.11	225.00	0.045

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	47.655 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	42.250 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	55.249 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	40.324 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	46.475 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	46.475 bars
App 27-4	= M.A.W.P.	36.658 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 36.658 • 1.0
 = 47.655 bars

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom D end	180.4	234.0	0.771	48.04
Shell	123.5	234.0	0.528	48.02
Top D end	134.5	234.0	0.575	47.73

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_2"	Nozzle	138.00	138.00	1.000
N1_16"	Nozzle	138.00	138.00	1.000
N5_8"	Nozzle	138.00	138.00	1.000
K1A_3"	Nozzle	138.00	138.00	1.000
K1B_3"	Nozzle	138.00	138.00	1.000
K2A_2"	Nozzle	138.00	138.00	1.000
K2B_2"	Nozzle	138.00	138.00	1.000
M1_24"	Nozzle	138.00	138.00	1.000
N2_16"	Nozzle	138.00	138.00	1.000
N4_2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom D end	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Top D end	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N3_2"	10.16	225.00	0.045
N1_16"	84.19	225.00	0.374
N5_8"	43.54	225.00	0.193
K1A_3"	10.16	225.00	0.045
K1B_3"	13.79	225.00	0.061
K2A_2"	10.16	225.00	0.045
K2B_2"	10.16	225.00	0.045
M1_24"	124.85	225.00	0.555



N2_16"	83.69	225.00	0.372
N4__2"	10.10	225.00	0.045

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	2800	...	...	No Calc	No Calc
20	30	2800	...	...	No Calc	No Calc
30	40	No Calc	928	12	0.001796	101.266
40	50	3150	940	18	0.00096559	84.2159
50	60	No Calc	936	16	0.0023742	107.442

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	14	4.40276	1.03419	14.5488
40	50	20	6.98392	1.03419	21.5007
50	60	18	4.4235	1.03419	20.4056

Minimum

15

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	2800	No Calc	No Calc	No Calc
20	30	2800	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	3150	145024	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	528.903	...	468.421	...	37.0232
20	30	385.537	...	351.935	...	26.9876
30	40	142.375	127257	124.578	128819	9.96625
40	50	1299.26	1845231	1171.87	1861669	90.9481
50	60	198.177	127257	180.161	128819	13.8724
Total		2554	2099745.50	2296	2119307.50	178

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
30	Liqd	125.396	...	-112.5	...	Liquid 30
30	Nozl	14.546	...	-368	...	N3_2"
40	Liqd	438.887	...	350	...	Liquid: 40
40	Nozl	187.505	575.006	1450	...	N1_16"
40	Nozl	64.7481	550	2350	...	N5_8"
40	Nozl	10.4632	553.187	750	...	K1A_3"
40	Nozl	17.6591	...	750	-550	K1B_3"
40	Nozl	11.0703	550	150	...	K2A_2"
40	Nozl	11.0703	...	150	-550	K2B_2"
40	Nozl	762.569	-516.688	800	...	M1_24"
40	Wght	3	400	150	...	Rung
40	Wght	3	-400	300	...	Rung
40	Wght	3	-400	1200	...	Rung
40	Wght	3	-400	1450	...	Rung
40	Wght	400	-450	2600	...	Pipe Davit
40	Wght	150	...	1450	...	Lifting Wt
40	Wght	400	...	1600	...	Clip Wt
40	Wght	50	-950	800	...	Manway davit
40	Wght	40	...	2500	...	Demister WT
40	Wght	115	...	1450	...	N1_Deflector
40	Wght	5	...	2500	...	Demister Suppor
50	Plat	2674.28	...	600	...	PLAT 1
50	Nozl	236.183	...	497.242	...	N2_16"
50	Nozl	11.1995	400	244.077	...	N4_2"

Total Weight of Each Detail Type:

Platforms	2674.3
Liquid	564.3
Nozzles	1327.0
Weights	1172.0

Sum of the Detail Weights 5737.6 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	2733.0	2733.0	2733.0	2733.0	2733.0	2733.0
Nozzles	1327.0	1327.0	1327.0	1327.0	1327.0	1327.0
Wld Weights	1017.0	1017.0	1017.0	1017.0	1017.0	1017.0
Platforms	...	...	2674.3	2674.3	2674.3	2674.3
Empty Weights	...	...	...	155.0	155.0	...
Test Weights	...	155.0	...	...	...	...
Ope Weights	...	...	...	...	...	155.0
Ope. Liquid	...	...	...	...	...	564.3
Test Liquid	...	2098.5	...	...	...	...



Totals	5077.1	7330.5	7751.3	7906.3	7906.3	8470.6
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Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	5077.1 kg.
Shop Test Wt.	- Fabricated weight + water (full)	7330.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	7751.3 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	7906.3 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	7906.3 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	8470.6 kg.
Field Test Wt.	- Empty weight + test medium (full)	10004.8 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	5297.0 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	5837.4 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	5404.8 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	52477.2	51572.4
20	30	29154	28651.3
30	40	10879.7	10194
40	50	85639.8	81995.6
50	60	11142.6	10194

Total	189293.3	182607.3 cm2
Total	203.8	196.6 ft²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	565.926	565.926	565.926	...	565.926
20	30	412.525	412.525	412.525	...	412.525
30	40	166.887	292.284	294.067	...	166.887
40	50	3627.29	4066.18	5471.4	468.615	3627.29
50	60	3133.72	3133.72	3260.89	4.47988	3133.72

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	7906.35	8470.63	10004.8
20	30	7340.42	7904.7	9438.88
30	40	6927.9	7492.18	9026.36
40	50	6761.01	7199.89	8732.29
50	60	3133.72	3133.72	3260.89

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.



Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	473.095	473.095	473.095
20	30	473.095	473.095	473.095
30	40	473.095	473.095	473.095
40	50	473.095	473.095	473.095
50	60	4.47988	4.47988	4.47988

Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N3_2"	46.06	51.00	120	300	GR 1.1	...	...	...	...
N1_16"	46.06	51.00	120	300	GR 1.1	21.334	46.060	46.060	46.060
N5_8"	46.06	51.00	120	300	GR 1.1	30.485	38.606	46.060	38.606
K1A_3"	46.06	51.00	120	300	GR 1.1	...	...	...	...
K1B_3"	46.06	51.00	120	300	GR 1.1	...	...	...	...
K2A_2"	46.06	51.00	120	300	GR 1.1	...	...	...	...
K2B_2"	46.06	51.00	120	300	GR 1.1	...	...	...	...
M1_24"	46.06	51.00	120	300	GR 1.1	...	...	...	...
N2_16"	46.06	51.00	120	300	GR 1.1	21.333	46.060	46.060	46.060
N4_2"	46.06	51.00	120	300	GR 1.1	...	...	...	...
Min Rating	38.606	51.000, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N1_16"	442.976	0.50	3424	2611	36.658	46.060	Pass
N5_8"	248.539	0.50	856	1305	36.658	38.606	Pass
N2_16"	442.976	0.50	3424	2611	36.658	46.060	Pass

Evaluating Flange on Nozzle [N116"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(3424) + 4 \cdot \text{abs}(2611) \cdot 442.976 \leq \pi \cdot 442.976^3 \cdot [(46.060 - 36.658) + .500 \cdot 46.060]$$

$$= 0.59412E+05 \leq 0.90319E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N116" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(46.060, 0.500 \cdot 46.060 + 46.060 - 59411.6 / (\pi \cdot 442.976^3))$$

$$= \min(46.060, 47.756)$$

$$= 46.060 \text{ bars}$$

Evaluating Flange on Nozzle [N58"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(856) + 4 \cdot \text{abs}(1305) \cdot 248.539 \leq \pi \cdot 248.539^3 \cdot [(46.060 - 36.658) + .500 \cdot 46.060]$$

$$= 0.14994E+05 \leq 0.15952E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N58" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(46.060, 0.500 \cdot 46.060 + 46.060 - 14994.1 / (\pi \cdot 248.539^3))$$

$$= \min(46.060, 38.606)$$

$$= 38.606 \text{ bars}$$

Evaluating Flange on Nozzle [N216"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$$

$$= 16(3424) + 4 \cdot \text{abs}(2611) \cdot 442.976 \leq \pi \cdot 442.976^3 \cdot [(46.060 - 36.658) + .500 \cdot 46.060]$$

$$= 0.59408E+05 \leq 0.90319E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N216" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$



= min(46.060, $0.500 \cdot 46.060 + 46.060 - 59408.2/(\pi \cdot 442.976^3)$)
= min(46.060, 47.758)
= 46.060 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 5.61408 Hz.

The Natural Frequency for the Vessel (Ope...) is 5.47887 Hz.

The Natural Frequency for the Vessel (Filled) is 5.13393 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.179 \cdot 0.962) / (1 + 1.7 \cdot 3.4 \cdot 0.179)))$
 $= \min(0.85, 0.907)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	5.479 Hz
Natural Frequency of Vessel (Empty)	5.614 Hz
Natural Frequency of Vessel (Test)	5.134 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.593
Structure Height to Diameter ratio	6.580
Height to top of Structure	6043.000 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$; $Z_g = 589788$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 900.0 + 300.0 = 1200.0$ mm.
 $= 3.937$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (3.937 ft.) < 15 ft.
 $= 2.41(15 / Z_g)^{2/\alpha}$
 $= 2.41(15 / 1935.0)^{2/11.5}$
 $= 1.035$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[K_{zt}]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.035 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$$

$$= 33.1 \text{ psf } [161.849] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.149 \cdot 0.85 \cdot 0.593 \cdot 31.465$$

$$= 525.7 \text{ lbs. } [238.5] \text{ Kg}$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Skirt 1	1200.0	0.000	0.000	0.000	1.035	1.000	161.849
Skirt 2	2600.0	0.000	0.000	0.000	1.035	1.000	161.849
Bottom D end	3125.0	0.000	0.000	0.000	1.035	1.000	161.849
Shell	4600.0	0.000	0.000	0.000	1.036	1.000	162.021
Top D end	6176.5	0.000	0.000	0.000	1.091	1.000	170.542

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm.
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm.
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm.
- Lc - Total length of conical section(s) of vessel, mm.
- tb - Uncorroded plate thickness at bottom of vessel, mm.
- V30 - Design Wind Speed provided by user, Km/hr
- Vc - Critical wind velocity, Km/hr
- Vw - Maximum wind speed at top of structure, Km/hr
- W - Total corroded weight of structure, Kg
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
- Z - Maximum amplitude of vibration at top of vessel, mm.
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.320E-06 (High); 0.320E-06 < 0.400E-06 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0.0/5800.0 = 0.0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[3.06/19.03^2] \cdot 10^4 = 84.633$ [Geometry Violation]

Compute the vibration possibility. If Vp > 0.400E-06 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 8195 / (5800.0 \cdot 939.931^2)$$

$$= 0.15994E-05$$

Since Vp is > 0.400E-06 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area	Elevation	Pressure	Force	Cf
----	-----------	-----------	----------	-------	----



	cm2	mm.	Kgs/m2	Kgf	
PLAT 1	119955.00	6650.00	171.48	2057.02	2.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf
Rung	0.00	3300.00	161.87	0.00
Rung	0.00	3450.00	161.89	0.00
Rung	0.00	4350.00	161.99	0.00
Rung	0.00	4600.00	162.02	0.00
Pipe Davit	0.00	5750.00	168.24	0.00
Lifting Wt	0.00	4600.00	162.02	0.00
Clip Wt	0.00	4750.00	162.83	0.00
Manway davit	0.00	3950.00	161.95	0.00
Demister WT	0.00	5650.00	167.70	0.00
N1Deflector	0.00	4600.00	162.02	0.00
Demister Supp	0.00	5650.00	167.70	0.00

The Natural Frequency for the Vessel (Ope...) is 5.47887 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	1200	1624	29232	161.849	262.323
20	30	2600	1624	16240	161.849	145.735
30	40	3125	1624	812	161.849	7.28676
40	50	4600	1645	47705	162.021	428.552
50	60	6176.53	1638	3945.15	170.542	2473.48

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		1.500	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	0.000	
Force Factor		0.000	
Consider Vertical Acceleration		No	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(19.767^{10.75})$
 = 0.187 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/5.479)
 = 0.183

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.187$, 0.183) per 12.8.2
 = 0.183

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.217 / (1.5 / 1.5)$
 = 0.217

Maximum value of C_s per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.093 / (0.183(1.5 / 1.5))$
 = 0.508

Minimum value of C_s per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
 = $\max((0.044 \cdot 0.2173 \cdot 1.5), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.2173 \cdot 8470.6$
 = 1840.950 Kgf

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} \cdot V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$ and k is an exponent which is related

to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 5.47887 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	900	565.926	24.7628	24.5674	...
20	30	2300	412.525	46.1291	45.7652	...
30	40	2825	292.284	40.1439	22.7405	...
40	50	4300	4066.18	850.065	752.33	...
50	60	5775	3133.72	879.85	872.909	...

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	900	3317.38	1840.95	16970.5	9067.71
20	30	2300	3055.05	1816.19	11235.2	5776.22
30	40	2825	2909.32	1770.06	8252.91	3983.07
40	50	4300	2902.03	1729.91	8107.63	3895.56
50	60	5775	2473.48	879.85	312.982	111.332

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	3317.38	0.14542	0.54652	0.0005686
20	30	3055.05	0.86095	1.23	0.00078719
30	40	2909.32	1.24972	1.2695	0.00079281
40	50	2902.03	2.48266	3.77292	0.00089983
50	60	2473.48	3.79542	3.81791	0.00089984

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	159.727	998.293
20	30	159.727	998.293
30	40	159.727	998.293
40	50	161.792	1011.2
50	60	161.104	1006.9

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 29.000 Actual deflection : 3.818 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	231.221	187.867	5272642	4290841
20	30	231.221	187.867	5272642	4290841
30	40	401.999	345.324	9049216	7806999
40	50	578.053	521.378	13018484	11792124
50	60	519.117	462.442	11689121	10457521

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	91.0784	154.437	-120.838	-120.838
20	30	91.0784	154.437	-120.838	-120.838
30	40	165.597	280.795	-135.833	-135.833
40	50	165.597	280.795	-142.651	-142.651
50	60	165.597	280.795	-140.906	-140.906

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	60.5619	-2.02575	88.803
40	50	40.1579	-1.37663	58.8843
50	60	45.2589	-1.5389	66.3639

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-4.12718	-4.42174	-5.2226	1.08125	1.08125
20	30	-3.83176	-4.12633	-4.92718	1.08125	1.08125
30	40	-1.96745	-1.96745	-1.96745	0.59427	0.59427
40	50	-1.27171	-1.27171	-1.27171	0.39344	0.39344
50	60	-0.66455	-0.66455	-0.66455	0.004201	0.004201

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	1.08125	38.7857	20.7241	12.7993	...
20	30	1.08125	25.6777	13.2015	8.47365	...
30	40	0.59427	10.3668	5.00326	3.42103	...
40	50	0.39344	6.74251	3.23965	2.22503	...
50	60	0.004201	0.2935	0.1044	0.096855	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	19.5557
20	30	...	...	...	12.5372
30	40	...	...	...	4.79174
40	50	...	...	...	3.10433
50	60	...	...	...	0.10358

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
10	20	...	...
20	30	...	...
30	40	...	...

40	50	...	...
50	60	...	...

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm^2	Earthquake For/Mom Corroded N./mm^2	Wind For/Mom No Corrosion N./mm^2	Earthquake For/Mom No Corrosion N./mm^2
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm^2
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	35.74	91.08	-43.99	120.84	0.3924	0.3641
20	22.93	91.08	-30.59	120.84	0.2517	0.2532
30	8.99	165.60	-12.93	135.83	0.0543	0.0952
40	5.86	165.60	-8.41	142.65	0.0354	0.0589
50		165.60	-0.96	140.91		0.0068

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	16.51	91.08	-24.76	120.84	0.1813	0.2049
20	9.79	91.08	-17.45	120.84	0.1075	0.1444
30	3.42	165.60	-7.35	135.83	0.0206	0.0541
40	2.23	165.60	-4.77	142.65	0.0134	0.0334
50		165.60	-0.77	140.91		0.0055

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	35.45	91.08	-44.29	120.84	0.3892	0.3665
20	22.63	91.08	-30.89	120.84	0.2485	0.2556
30	8.99	165.60	-12.93	135.83	0.0543	0.0952
40	5.86	165.60	-8.41	142.65	0.0354	0.0589



50 165.60 -0.96 140.91 0.0068

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	17.38	91.08	-26.23	120.84	0.1909	0.2170
20	10.16	91.08	-18.41	120.84	0.1115	0.1523
30	3.63	165.60	-7.56	135.83	0.0219	0.0557
40	2.36	165.60	-4.90	142.65	0.0143	0.0344
50		165.60	-0.77	140.91		0.0055

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.66	154.44	-19.10	120.84	0.0561	0.1581
20	4.63	154.44	-14.48	120.84	0.0300	0.1198
30	2.05	280.79	-5.98	135.83	0.0073	0.0440
40	1.35	280.79	-3.89	142.65	0.0048	0.0273
50		280.79	-0.77	140.91		0.0054

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.44	-6.30	120.84		0.0522
20		154.44	-6.01	120.84		0.0497
30		280.79	-2.56	135.83		0.0189
40		280.79	-1.67	142.65		0.0117
50		280.79	-0.67	140.91		0.0047

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	35.45	91.08	-44.29	120.84	0.3892	0.3665
20	22.63	91.08	-30.89	120.84	0.2485	0.2556
30	69.56	165.60		135.83	0.4200	
40	46.02	165.60		142.65	0.2779	
50	44.89	165.60		140.91	0.2711	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	17.38	91.08	-26.23	120.84	0.1909	0.2170
20	10.16	91.08	-18.41	120.84	0.1115	0.1523
30	64.19	165.60		135.83	0.3876	
40	42.52	165.60		142.65	0.2568	
50	44.70	165.60		140.91	0.2700	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	35.45	91.08	-44.29	120.84	0.3892	0.3665
20	22.63	91.08	-30.89	120.84	0.2485	0.2556
30	6.97	165.60	-14.95	135.83	0.0421	0.1101
40	4.49	165.60	-9.78	142.65	0.0271	0.0686
50		165.60	-2.50	140.91		0.0178

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	17.38	91.08	-26.23	120.84	0.1909	0.2170
20	10.16	91.08	-18.41	120.84	0.1115	0.1523
30	1.60	165.60	-9.59	135.83	0.0097	0.0706
40	0.98	165.60	-6.28	142.65	0.0059	0.0440



50 165.60 -2.31 140.91 0.0164

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.66	154.44	-19.10	120.84	0.0561	0.1581
20	4.63	154.44	-14.48	120.84	0.0300	0.1198
30	90.85	280.79		135.83	0.3235	
40	60.23	280.79		142.65	0.2145	
50	65.80	280.79		140.91	0.2343	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.44	-6.30	120.84		0.0522
20		154.44	-6.01	120.84		0.0497
30	87.43	280.79		135.83	0.3114	
40	58.01	280.79		142.65	0.2066	
50	65.70	280.79		140.91	0.2340	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-5.21	120.84		0.0431
20		91.08	-4.91	120.84		0.0407
30	59.19	165.60		135.83	0.3574	
40	39.28	165.60		142.65	0.2372	
50	44.60	165.60		140.91	0.2693	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-3.59	120.84		0.0297
20		91.08	-3.35	120.84		0.0277
30	58.87	165.60		135.83	0.3555	
40	39.01	165.60		142.65	0.2356	
50	44.67	165.60		140.91	0.2697	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-5.50	120.84		0.0455
20		91.08	-5.21	120.84		0.0431
30	59.19	165.60		135.83	0.3574	
40	39.28	165.60		142.65	0.2372	
50	44.60	165.60		140.91	0.2693	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-5.21	120.84		0.0431
20		91.08	-4.91	120.84		0.0407
30	59.19	165.60		135.83	0.3574	
40	39.28	165.60		142.65	0.2372	
50	44.60	165.60		140.91	0.2693	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-5.50	120.84		0.0455
20		91.08	-5.21	120.84		0.0431
30		165.60	-2.56	135.83		0.0189
40		165.60	-1.67	142.65		0.0117



50 165.60 -0.67 140.91 0.0047

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		75.90	-5.50	100.70		0.0546
20		75.90	-5.21	100.70		0.0517
30	59.19	138.00		113.19	0.4289	
40	39.28	138.00		118.88	0.2846	
50	44.60	138.00		117.42	0.3232	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		75.90	-5.50	100.70		0.0546
20		75.90	-5.21	100.70		0.0517
30		138.00	-4.59	113.19		0.0405
40		138.00	-3.04	118.88		0.0256
50		138.00	-2.21	117.42		0.0188

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		91.08	-5.50	120.84		0.0455
20		91.08	-5.21	120.84		0.0431
30	59.19	165.60		135.83	0.3574	
40	39.28	165.60		142.65	0.2372	
50	44.60	165.60		140.91	0.2693	

Absolute Maximum of the all of the Stress Ratio's 0.4289

Governing Element: Bottom D end
 Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt 1	...	-4.4	1.1	...	...	+ Tens
Skirt 1	...	-4.4	-1.1	...	-5.5	- Comp
Skirt 2	...	-4.1	1.1	...	...	+ Tens
Skirt 2	...	-4.1	-1.1	...	-5.2	- Comp
Bottom D end	60.6	-2.0	0.6	...	59.2	+ Tens
Bottom D end	60.6	-2.0	-0.6	...	...	- Comp
Shell	40.2	-1.3	0.4	...	39.3	+ Tens
Shell	40.2	-1.3	-0.4	...	...	- Comp
Top D end	45.3	-0.7	...	...	44.6	+ Tens
Top D end	45.3	-0.7	...	...	...	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt 2

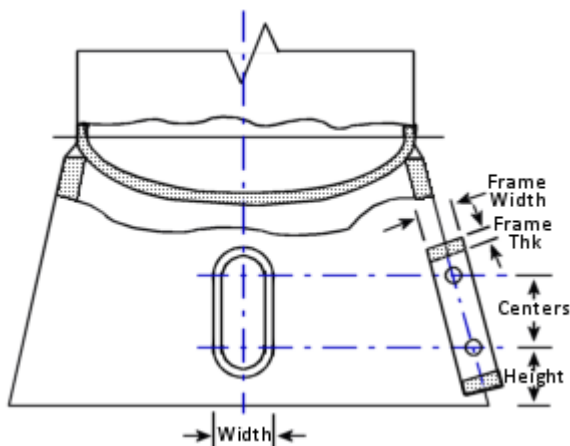
Total vessel (empty) weight	Fg	6991.905 Kgf
Head Weight below section 2-2	DeltaFg	166.887 Kgf
Weight of vessel contents (liquid)	Ff	564.283 Kgf
Vessel Gross Weight	Fg + Ff	7556.189 Kgf
Skirt Mean Diameter	Dz	920.000 mm.
Skirt Analysis Thickness (Corroded)	ez	6.500 mm.
Skirt Height	H	1000.000 mm.
Additional Axial Force at section 1-1	F1	0.000 Kgf
Moment at section 1-1	M1	8252.914 Kg-m.
Applied Moment - Bottom of Skirt	Mb	11235.159 Kg-m.
Weight at Bottom of Skirt	W	7904.703 Kgf

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	916.000 mm.
Head Corroded Thickness	eb	12.000 mm.

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 227.0 \text{ mm.} \\
 K &= D_i / (2 * h_i) = 904.0 / (2 * 227.0) = 1.9912 \\
 R &= D_i (0.44 * K + 0.02) = 904.0 * (0.44 * 1.991 + 0.02) = 810.0955 \text{ mm.} \\
 r &= D_i ((0.5 / K) - 0.08) = 904.0 * ((0.5 / 1.991) - 0.08) = 154.68 \text{ mm.}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	198763.391 cm ⁴
Moment of Inertia	Iyy =	198763.391 cm ⁴
Section Modulus	W =	4320946.500 mm. ³
Section Metal Area	A =	187.867 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(198763.391, 198763.391) / 187.867)} \\
 &= 325.269 \text{ mm.}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:



$$= rg / L$$

$$= 325.269/1000.0$$

$$= 0.325$$

Correct the Moment [M4]:

top of skirt is the critical end

Controlling Moment [M4] = 8252.914 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -8252.914/17283786 - 7904.703/187.86725$$

$$= -8.809 \text{ N./mm}^2$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +8252.914/17283786 - 7904.703/187.86725$$

$$= 0.556 \text{ N./mm}^2$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F1 - Fg - Ff + 4 \cdot M1 / Dz$$

$$= 0.0 - 6991.905 - 564.283 + 4 \cdot 8252.914/920.0$$

$$= 28325.3 \text{ Kgf}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F1 - Fg - Ff - 4 \cdot M1 / Dz$$

$$= 0.0 - 6991.905 - 564.283 - 4 \cdot 8252.914/920.0$$

$$= -43437.7 \text{ Kgf}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (28325.334 + 166.887 + 564.283) / (\pi \cdot 916.0 \cdot 12.0) + 32.5 \cdot 916.0 / (4 \cdot 12.0)$$

$$= 70.28 \text{ N./mm}^2$$

Stress [σ_{1qm}]:

$$= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-43437.715 + 166.887 + 564.283) / (\pi \cdot 916.0 \cdot 12.0) + 32.5 \cdot 916.0 / (4 \cdot 12.0)$$

$$= 49.90 \text{ N./mm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (564.283 + 166.887) / (\pi \cdot 916.0 \cdot 12.0) + 32.5 \cdot 916.0 / (4 \cdot 12.0)$$

$$= 62.23 \text{ N./mm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= Fzp / (\pi \cdot Dz \cdot ez)$$

$$= 28325.334 / (\pi \cdot 920.0 \cdot 6.5)$$

$$= 14.79 \text{ N./mm}^2$$

Stress [σ_{3qm}]:

$$= Fzq / (\pi \cdot Dz \cdot ez)$$

$$= -43437.715 / (\pi \cdot 920.0 \cdot 6.5)$$

$$= -22.67 \text{ N./mm}^2$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(Db + eb - Dz + ez) / (r + eb)$$

$$= 1 - 0.5(916.0 + 12.0 - 920.0 + 6.5) / (154.68 + 12.0)$$

$$= 0.9565$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.957) = 16.961 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(12.0^2 + 6.5^2 + 2 \cdot 12.0 \cdot 6.5 \cdot 0.957)}$$

$$= 9.158 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 9.158 \cdot 28325.3 = 259.40 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 9.158 \cdot -43437.7 = -397.80 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2$$

$$= 0.63 - 0.057 \left(\frac{12.0}{6.5} \right)^2$$

$$= 0.436$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.436 \cdot 6 \cdot 259.404 / (\pi \cdot 916.0 \cdot 12.0^2)$$

$$= 16.05 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.436 \cdot 6 \cdot -397.804 / (\pi \cdot 916.0 \cdot 12.0^2)$$

$$= -24.61 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.436 \cdot 6 \cdot 259.404 / (\pi \cdot 920.0 \cdot 6.5^2)$$

$$= 54.46 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.436 \cdot 6 \cdot -397.804 / (\pi \cdot 920.0 \cdot 6.5^2)$$

$$= -83.52 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 0.472/36.063 = 0.0131$$

$$y = 125 \cdot eb/D_b = 125 \cdot 12.0/916.0 = 1.6376$$

As this an elliptical head, it will be treated as a Korbbogen type head

Geometric value [α]:

$$= 4.2 - 0.2 \cdot y = 4.2 - 0.2 \cdot 1.638 = 3.8725$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b \left(\left(\frac{y}{y_A} \right) \cdot \alpha - 1 \right) / (4 \cdot eb)$$

$$= (32.5 + 0.0) \cdot 916.0 \left(\left(\frac{16.96}{40.0} \right) \cdot 3.872 - 1 \right) / (4 \cdot 12.0)$$

$$= 39.82 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 70.276 - 16.049 + 39.822$$

$$= 94.05 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 70.276 + 16.049 - 39.822$$

$$= 46.50 \text{ N./mm}^2$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 49.896 - 24.612 + 39.822$$



$$= 114.33 \text{ N./mm}^2$$

Stress [σ_{1qotot}]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= 49.896 + -24.612 - 39.822 \\ &= -14.54 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 62.232 + 16.049 + 39.822 \\ &= 118.10 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2potot}]:

$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 62.232 - 16.049 - 39.822 \\ &= 6.36 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2qitot}]:

$$\begin{aligned} &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\ &= 62.232 + -24.612 + 39.822 \\ &= 77.44 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{2qotot}]:

$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 62.232 - -24.612 - 39.822 \\ &= 47.02 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{3pitot}]:

$$\begin{aligned} &= \sigma_{3pm} - \sigma_{3pb} \\ &= 14.786 - 54.463 \\ &= -39.68 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{3potot}]:

$$\begin{aligned} &= \sigma_{3pm} + \sigma_{3pb} \\ &= 14.786 + 54.463 \\ &= 69.25 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{3qitot}]:

$$\begin{aligned} &= \sigma_{3qm} - \sigma_{3qb} \\ &= -22.675 - -83.52 \\ &= 60.85 \text{ N./mm}^2 \end{aligned}$$

Stress [σ_{3qotot}]:

$$\begin{aligned} &= \sigma_{3qm} + \sigma_{3qb} \\ &= -22.675 + -83.52 \\ &= -106.19 \text{ N./mm}^2 \end{aligned}$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\ &= 137.997(3 - (1/1.5)) \cdot (70.276/137.997)^2 \\ &= 390.13 \text{ N./mm}^2 \end{aligned}$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\ &= 137.997(3 - (1/1.5)) \cdot (49.896/137.997)^2 \\ &= 401.97 \text{ N./mm}^2 \end{aligned}$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\ &= 137.997(3 - (1/1.5)) \cdot (62.232/137.997)^2 \\ &= 395.28 \text{ N./mm}^2 \end{aligned}$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\ &= 137.997(3 - (1/1.5)) \cdot (62.232/137.997)^2 \\ &= 395.28 \text{ N./mm}^2 \end{aligned}$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:



$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 137.997(3 - (1/1.5) \cdot (14.786/137.997)^2)$$

$$= 412.94 \text{ N./mm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 137.997(3 - (1/1.5) \cdot (-22.675/137.997)^2)$$

$$= 411.51 \text{ N./mm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm2)

Stress	Actual	Allowed	Result	Equation
Stot1pi	94.05	390.13	Pass	46
Stot1po	46.50	390.13	Pass	47
Stot1qi	114.33	401.97	Pass	48
Stot1qo	-14.54	401.97	Pass	49
Stot2pi	118.10	395.28	Pass	50
Stot2po	6.36	395.28	Pass	51
Stot2qi	77.44	395.28	Pass	52
Stot2qo	47.02	395.28	Pass	53
Stot3pi	-39.68	412.94	Pass	54
Stot3po	69.25	412.94	Pass	55
Stot3qi	60.85	411.51	Pass	56
Stot3qo	-106.19	411.51	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	6350.000 mm.
Center of Gravity of the Liquid	3086.111 mm.
Center of Gravity of the Nozzles	4274.259 mm.
Center of Gravity of the Added Weights (Operating)	4749.360 mm.
Center of Gravity of the Added Weights (Empty)	4749.360 mm.
Center of Gravity of Bare Shell New and Cold	3328.718 mm.
Center of Gravity of Bare Shell Corroded	3338.567 mm.
Vessel CG in the Operating Condition	4657.133 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	4719.942 mm.
Vessel CG in the Test Condition	4631.847 mm.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	928.0000	mm.
Skirt Thickness	STHK	8.0000	mm.
Skirt Internal Corrosion Allowance	SCA	0.7500	mm.
Skirt External Corrosion Allowance		0.7500	mm.
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	26.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-283 C	
Basing Ope. All. Stress	BASOPE, f4	108.00	N./mm ²
Basing Yield Stress		207.00	N./mm ²
Inside Diameter of Basing	DI, D4	840.0000	mm.
Outside Diameter of Basing	DOU, D5	1146.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	30.0000	mm.
Bolt Corrosion Allowance	BCA	1.5000	mm.
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	149.94	N./mm ²
Number of Bolts	RN, nb	8	
Diameter of Bolt Circle	DC, DBC	1044.0000	mm.
Bolt Allowable Shear Stress		99.930	N./mm ²
Ultimate Comp. Strength of Concrete	FPC	35.0	N./mm ²
Allowable Comp. Strength of Concrete	FC	15.8	N./mm ²
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	12.0000	mm.
Width of Gussets at Top Plate	TWDT	109.0000	mm.
Width of Gussets at Base Plate	BWDT	109.0000	mm.
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	207.0	N./mm ²
Height of Chair	h1	242.0000	mm.
Distance between Gussets	RG, b6	102.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	51.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	29.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	109.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	51.0000	mm.
External Corrosion Allowance	CA	2.0000	mm.
Dead Weight of Vessel	DW	7906.3	Kgf
Operating Weight of Vessel	ROW	8470.6	Kgf
Earthquake Moment on Basing	EQMOM	9297.6	Kg-m.
Wind Moment on Basing	WIMOM	17200.4	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 9298/1044.0) - 8471) / 8$$

$$= 3394.0 \text{ Kgf}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 17200/1044.0) - 7906) / 8$$

$$= 7249.3 \text{ Kgf}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/1044.0) - 8471) / 8$$

$$= -1058.8 \text{ Kgf}$$

Governing Bolt Load Condition, Wind + Dead Weight Condition:

Area Available in 1 Bolt (corroded)	Abss :	3.9078	cm ²
Area Available in all the Bolts	Abss * RN :	31.2625	cm ²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.243	0.121	1.348	2.567	0.450	0.778	16598.4	24504.7	8384.1
4	0.182	0.152	1.160	2.701	0.463	0.774	16556.4	24462.7	7949.3
6	0.167	0.174	1.108	2.734	0.466	0.773	16550.3	24456.7	7848.0
8	0.171	0.173	1.121	2.726	0.465	0.773	16551.7	24458.0	7873.1
10	0.172	0.172	1.125	2.724	0.465	0.773	16552.0	24458.4	7879.4
12	0.172	0.172	1.126	2.723	0.465	0.773	16552.1	24458.5	7880.9

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 16552.1 / (0.953 \cdot 1044.0 \cdot 2.723)$$

$$= 119.795 \text{ N./mm}^2, \text{ Should be } \leq 149.9 \text{ N./mm}^2$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 8 \cdot 3.908 / (\pi \cdot 1044.0)$$

$$= 0.953 \text{ mm.}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc / [(2kd + t3) / (2kd)]$$

$$= 2.529 / [(2 \cdot 0.1718 \cdot 1044.0 + 153.0) / (2 \cdot 0.1718 \cdot 1044.0)]$$

$$= 3.608 \text{ N./mm}^2, \text{ Should be } \leq 15.8 \text{ N./mm}^2$$

Values for table 10.3, l = 109.0, b = 204.989, l/b = 0.531736

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(530.953, 1317.738) = 1317.738 \text{ Kgf}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow) + Ca}$$

$$= \sqrt{(6 \cdot 1317.738 / 136.6) + 2.0}$$

$$= 25.823 \text{ mm.}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 109.0, b = 102.0, b/l = 0.93578

As b/l (0.936) is less than 1, inverting b/l = 1.069.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3 (\ln((2 \cdot l \sin(\pi \cdot a / l) / (\pi \cdot g))) + 1)] - [(0.7 - g2) P / (4 \pi)]$$

$$= 5974.8 / (4 \cdot \pi) [1.3 (\ln((2 \cdot 109.0 \cdot \sin(\pi \cdot 58.0 / 109.0)) / (\pi \cdot 25.0))) + 1] - [(0.7 - 0.128) \cdot 5974.8 / (4 \cdot \pi)]$$

$$= 831.5435 \text{ Kgf}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot Abs(Mo) / Fb) + Ca}$$

$$= \sqrt{(6 \cdot Abs(831.54) / 138.0) + 2.0}$$

$$= 20.8299 \text{ mm.}$$



Required Gusset Plate Thickness [tg]:
 $= P / (\text{Stress Term} \cdot l_{avg}) + Ca$
 $= 5974.8 / (124.11 \cdot 109.0) + 2.0$
 $= 11.525 \text{ (not less than } 9.525 + 2.0 \text{) mm.}$

Bolt spacing [m]:
 $= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$
 $= \pi \cdot 1044.0 / 8 = 409.978 \text{ mm.}$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:
 $= 1.76(P \cdot a / (m \cdot (h + t_{ba}) \cdot 1.5 \cdot Sktope))^{2/3} \cdot r^{1/3} + Ca$
 $= 1.76(5975 \cdot 58.0 / (409.98 \cdot 213.0 \cdot 207))^{2/3} \cdot 464.01^{1/3} + Ca$
 $= 4.472 + 1.5 = 5.972 \text{ mm.}$

Shear Stress in a Single Bolt [tb]:
 $= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 3317.4 / (3.908 \cdot 8)$
 $= 10.4 \text{ N./mm}^2. \text{ Must be less than } 99.9 \text{ N./mm}^2.$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	25.8233	mm.
Actual Basing Thickness as entered by user	26.0000	mm.
Required Thickness of Chair Cap	20.8299	mm.
Actual Top Ring Thickness as entered by user	29.0000	mm.
Required Gusset thickness, + CA	11.5250	mm.
Actual Gusset Thickness as entered by user	12.0000	mm.
Required Thickness of Skirt for Local Stress	5.9717	mm.
Given Thickness of Skirt	8.0000	mm.
Required Gusset Height to meet local stress	111.0433	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:
 $= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$
 $= 44.289 \cdot (8.0 - 1.5)$
 $= 29.35 \text{ Kg/mm.}$

Results for Computed Minimum Basing Weld Size [BWeld]:
 $= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 29 / [(0.4 \cdot 207) \cdot 2 \cdot 0.707]$
 $= 2.459 \text{ mm.}$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:
 $= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (Hg + Tta))$
 $= 5974.8 / (182.0 + 2 \cdot (187.0 + 29.0))$
 $= 9.731 \text{ Kg/mm.}$

Horizontal Plate Load [Wh]:
 $= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (Hg + Tta) + 0.6667 \cdot (Hg + Tta)^2)$
 $= 5974.8 \cdot 58.0 / (182.0 \cdot (216.0) + 0.6667 \cdot (216.0)^2)$
 $= 4.921 \text{ Kg/mm.}$

Resultant Weld Load [Wr]:
 $= \sqrt{ Wv^2 + Wh^2 }$
 $= \sqrt{ 9.73^2 + 4.92^2 }$
 $= 10.905 \text{ Kg/mm.}$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:
 $= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 10.9 / [(0.4 \cdot 207) \cdot 2 \cdot 0.707]$
 $= 0.913 \text{ mm.}$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:
= Bolt Load / (2 • TopWth)
= 5974.8/(2 • 109.0)
= 27.407 Kgf/mm.

Weld Load [Wh]:
= Bolt Load • e / (2 • Hgt • TopWth)
= 5974.8 • 58.0/(2 • 216.0 • 109.0)
= 7.359 Kgf/mm.

Resultant Weld Load [Wr]:
= $\sqrt{Wv^2 + Wh^2}$
= $\sqrt{27.41^2 + 7.36^2}$
= 28.378 Kgf/mm.

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:
= Wr / [(0.4 • Yield) • 2 • 0.707]
= 28.38/[(0.4 • 207) • 2 • 0.707]
= 2.377 mm.

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	4.7625	mm.
Required Gusset to Skirt Double Fillet Weld Size	4.7625	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	2.3771	mm.

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N32"

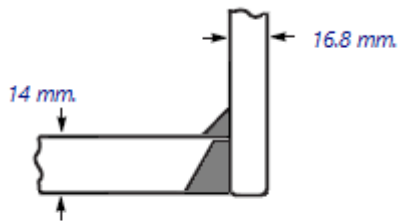
From: 30

Pressure for Reinforcement Calculations	P	36.752	bars
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	120	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	900.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	14.0000	mm.
Head Internal Corrosion Allowance	cas	2.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		800.0	mm.
Centerline Distance to Tangent		1500.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N32"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 36.75 \cdot 904.0 \cdot 0.8962 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 36.75)$
 $= 10.8180$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27.4 \cdot (\exp([36.75 / (138.0 \cdot 1.0)] - 1))$
 $= 0.7396$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3871 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	109.6000	mm.
Parallel to Vessel Wall, opening length	d	54.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N32".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.7396	mm.
Wall Thickness per UG16(b),	tr16b	= 3.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 14.0000	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.0000	mm.
Wall Thickness, shell/head, external pressure	trb2	= 2.3368	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000	mm.
Wall Thickness per table UG-45	tb3	= 6.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.8, \max(14.0, 3.5)]$
 $= 6.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.7396, 6.8)$
 $= 6.8000$ mm.



Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 14.0, tr = 10.818, c = 2.0 mm., $E* = 1.0$
Thickness Ratio = $tr * E* / (tg - c) = 0.901$, Temp. Reduction = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -20 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $36.75/51.00 = 0.721$

Weld Size Calculations, Description: N32"

Intermediate Calc. for nozzle/shell Welds Tmin 14.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * Wo \text{ mm.}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0832 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 215.0832 mm.

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Input, Nozzle Desc: N116"

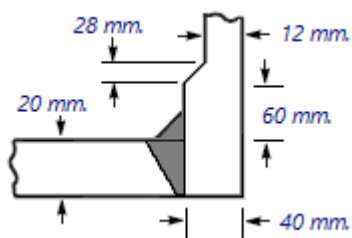
From: 40

Pressure for Reinforcement Calculations	P	36.658	bars
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	120	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	900.00	mm.
Design Length of Section	L	3150.0000	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1500.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	60.0000	mm.
Height of Beveled Transition	L'	28.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		40.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).





Reinforcement CALCULATION, Description: N116"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 16.000 in.
 Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 36.66 \cdot 452.0 / (138 \cdot 1.0 - 0.6 \cdot 36.66)$
 $= 12.2022$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 36.66 \cdot 205.2 / (138 \cdot 1.0 - 0.6 \cdot 36.66)$
 $= 5.5396$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.1027 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 28.0 / 0.5773$
 $= 48.4975$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

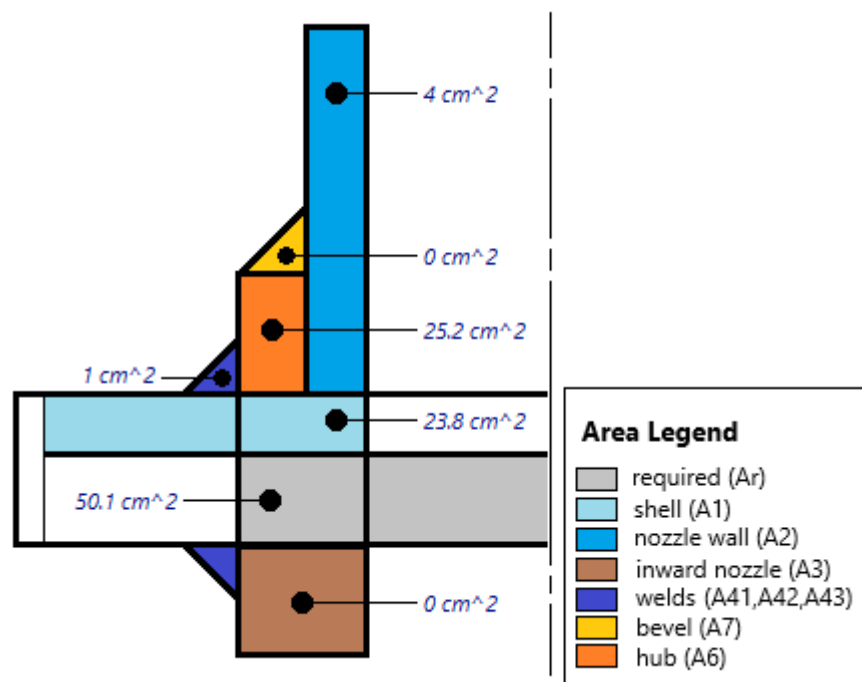
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	820.8000	mm.
Parallel to Vessel Wall, opening length	d	410.4000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	45.0000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	50.078	10.227	NA
Area in Shell	A1	23.794	53.418	NA
Area in Nozzle Wall	A2	4.014	8.008	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	25.200	25.200	NA
TOTAL AREA AVAILABLE	Atot	54.008	87.626	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (410.4 \cdot 12.2022 \cdot 1.0 + 2 \cdot 38.0 \cdot 12.2022 \cdot 1.0 \cdot (1 - 1.0))$
 $= 50.078 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 410.4(1.0 \cdot 18.0 - 1.0 \cdot 12.202) - 2 \cdot 38.0$
 $(1.0 \cdot 18.0 - 1.0 \cdot 12.2022) \cdot (1 - 1.0)$
 $= 23.794 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 45.0)(10.0 - 5.54)1.0$
 $= 4.014 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$



Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr2$$

$$= (2 \cdot \min(45.0, 250.0, 60.0)) \cdot (40.0 - 12.0) \cdot 1.0$$

$$= 25.200 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.5396 mm.
Wall Thickness per UG16(b),	tr16b = 3.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 14.2022 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 14.2022 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3389 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm.
Wall Thickness per table UG-45	tb3 = 10.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[10.331, \max(14.2022, 3.5)]$$

$$= 10.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(7.5396, 10.3312)$$

$$= 10.3312 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 59.5,	Allowable	: 138.0 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 285.5 N./mm2	Passed
Occasional	: 36.7,	Allowable	: 183.5 N./mm2	Passed
Shear	: 16.6,	Allowable	: 96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 12.0, tr = 5.54, c = 2.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.554, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-46 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 20.0, tr = 12.202, c = 2.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.678, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-22 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -22 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 36.66/51.00 = 0.719

Weld Size Calculations, Description: N116"



Intermediate Calc. for nozzle/shell Welds Tmin 18.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (50.0779 - 23.7941 + 2 \cdot 38.0 \cdot 1.0 \cdot (1.0 \cdot 18.0 - 12.2022)) \cdot 138)$
 $= 43185.87 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (29.2144 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 42516.35 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (29.2144 + 0.0 + 1. + (13.68)) \cdot 138$
 $= 61766.26 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (29.2144 + 0.0 + 1. + 0.0 + (13.68)) \cdot 138$
 $= 61766.26 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 486.4 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 52681. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 224.2) \cdot (40.0 - 2.0) \cdot 0.7 \cdot 138$
 $= 263639. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 486.4 \cdot (20.0 - 2.0) \cdot 0.74 \cdot 138$
 $= 143206. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (52681 + 263639) = 316320 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (52681 + 0 + 143206 + 0) = 195886 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (52681 + 143206 + 0) = 195886 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 316319 Kgf, must exceed W = 43185 Kgf	or W1 = 42516 Kgf
Path 2-2 = 195886 Kgf, must exceed W = 43185 Kgf	or W2 = 61766 Kgf
Path 3-3 = 195886 Kgf, must exceed W = 43185 Kgf	or W3 = 61766 Kgf



Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 38 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 71.3791 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 341.3791 mm.

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 354.99 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 410.40 mm.

Input Echo, WRC297 Item 1, Description: N116" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	900.000	mm.
Shell Thickness	20.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm2
Vessel Hot S.I. Allowable	Smh	138.00 N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	2.0000	mm.
Nozzle Diameter	406.400	mm.
Nozzle Thickness	40.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm2
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	36.66	bars
Radial Load	P	-2610.65	Kgf
Circumferential Shear	Vc	1957.99	Kgf
Longitudinal Shear	Vl	2610.65	Kgf
Circumferential Moment	Mc	2087.71	Kg-m.
Longitudinal Moment	Ml	2714.03	Kg-m.
Torsional Moment	Mt	3131.58	Kg-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

*Warning - The ratio, Dn/Dv (.517) is > 0.5
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$
 $= 486.4 + 2 \cdot 1.65 \cdot \sqrt{(461.0(20.0 - 2.0))}$
 $= 787.008 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	922.000	mm.
Nozzle Outside Diameter	(d) =	486.400	mm.
Vessel Thickness used	(T) =	18.000	mm.



Nozzle Thickness used (t) = 38.000 mm.
 T / t = 0.474
 d / t = 12.800

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.776$$

Nr/P = 0.056
 Mr/P = 0.083
 M0/P = 0.029
 N0/P = 0.084
 MrD/Mc = 0.257
 NrDL/Mc = 0.091
 M0d/Mc = 0.078
 N0DL/Mc = 0.090
 MrD/M1 = 0.083
 NrDL/M1 = 0.045
 M0D/M1 = 0.025
 N0DL/M1 = 0.097

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-16	-16	16	16
Outplane Bending (ML)	-24	24	24	-24
Normal Pressure Stress	90	93	90	93
Outplane Stress Summary	69	95	152	78

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	39	-39	39	-39
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-84	84	84	-84
Inplane Pressure Stress	45	45	45	45
Inplane Stress Summary	-2	86	180	-66

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	73	98	181	144

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
--	------------	------------	-------------	-------------



circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-60	60	60	-60
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	45	45	45	45
Outplane Stress Summary	-6	87	138	-11

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	39	-39	39	-39
Inplane Membrane (Mc)	-11	-11	11	11
Inplane Bending (Mc)	-200	200	200	-200
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	90	93	90	93
Inplane Stress Summary	-78	247	346	-129

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	78	247	346	130

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-16	-16	16	16
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	21	26	21	26
Hoop Stress Summary	12	16	44	49

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-4	-4	4	4



Axial Bending (ML)	-13	13	13	-13
Axial Pressure Stress	10	10	10	10

Axial Stress Summary	-1	14	35	-3
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	2	2	1	1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	14	18	44	52

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	21	26	21	26
Hoop Stress Summary	16	21	40	44

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-37	37	37	-37
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	10	10	10	10
Axial Stress Summary	-23	38	57	-27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	1	1	3	3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	40	38	58	72

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	90	93	90	93	90	93	90	93
Circ. Pl (SUS)	-10	-10	22	22	-7	-7	15	15
Circ. Q (SUS)	-11	11	37	-37	-161	161	239	-239
Long. Pm (SUS)	45	45	45	45	45	45	45	45
Long. Pl (SUS)	-3	-3	11	11	-5	-5	17	17
Long. Q (SUS)	-45	45	123	-123	-47	47	73	-73
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	90.0	93.0	90.0	93.0	90.0	93.0	90.0	93.0
Pm+Pl (SUS)	80.0	83.0	112.0	115.0	83.0	86.0	105.0	108.0
Pm+Pl+Q (Total)	72.7	96.6	179.3	145.1	78.1	247.1	344.1	131.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	93.00	138.00	Passed
Pm+Pl (SUS)	115.02	207.00	Passed
Pm+Pl+Q (TOTAL)	344.12	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	21	26	21	26	21	26	21	26
Circ. Pl (SUS)	-10	-10	22	22	-5	-5	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	-4	-4	4	4	-3	-3	3	3
Long. Q (SUS)	-8	8	18	-18	-32	32	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	21.0	26.0	21.0	26.0	21.0	26.0	21.0	26.0
Pm+Pl (SUS)	11.0	16.0	43.0	48.0	16.0	21.0	38.0	43.0
Pm+Pl+Q (Total)	13.6	17.2	43.4	52.2	41.2	39.2	55.2	72.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	26.00	138.00	Passed
Pm+Pl (SUS)	48.00	207.00	Passed
Pm+Pl+Q (TOTAL)	72.11	413.99	Passed



*Because only sustained loads were specified, the P_m+P_l+Q allowable was $3 * S_mh$.*

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Input, Nozzle Desc: N58"

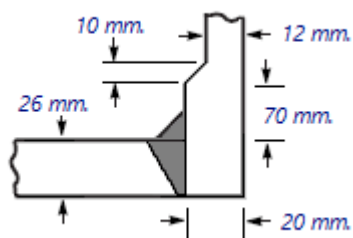
From: 40

Pressure for Reinforcement Calculations	P	36.658	bars
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	120	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	900.00	mm.
Design Length of Section	L	3150.0000	mm.
Shell Finished (Minimum) Thickness	t	26.0000	mm.
Shell Internal Corrosion Allowance	cas	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2400.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	10.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).





Reinforcement CALCULATION, Description: N58"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 8.000 in.
Actual Thickness Used in Calculation 0.472 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $36.66 \cdot 452.0 / (138 \cdot 1.0 - 0.6 \cdot 36.66)$
= 12.2022 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $36.66 \cdot 103.6 / (138 \cdot 1.0 - 0.6 \cdot 36.66)$
= 2.7968 mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.6891 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):
= 2.5 • Corroded Hub Thickness
= 2.5 • 18.0, less than the hub height, use (e-2)
= 45.0000 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

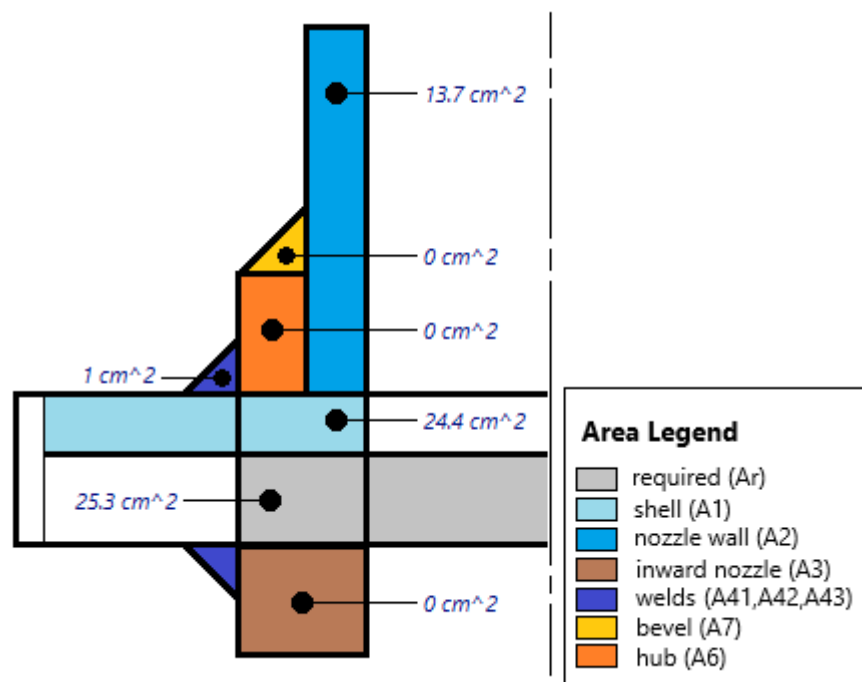
Parallel to Vessel Wall (Diameter Limit)	D1	414.4000	mm.
Parallel to Vessel Wall, opening length	d	207.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	45.0000	mm.

Hub Height was ≥ 2.5 times Hub Thickness, using sketch UG-40 (e-2).

Weld Strength Reduction Factor [fr1]:
= min(1, S_n/S_v)
= min(1, $138.0/138.0$)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n/S_v)
= min(1, $138.0/138.0$)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	25.283	5.163	NA
Area in Shell	A1	24.445	39.401	NA
Area in Nozzle Wall	A2	13.683	15.580	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	39.128	55.981	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (207.2 \cdot 12.2022 \cdot 1.0 + 2 \cdot 18.0 \cdot 12.2022 \cdot 1.0 \cdot (1 - 1.0))$
 $= 25.283 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 207.2(1.0 \cdot 24.0 - 1.0 \cdot 12.202) - 2 \cdot 18.0$
 $(1.0 \cdot 24.0 - 1.0 \cdot 12.2022) \cdot (1 - 1.0)$
 $= 24.445 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 45.0)(18.0 - 2.8)1.0$
 $= 13.683 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$



There are no hub area calculations because Figure UG-40 (e-2) was used.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 4.7968$ mm.
 Wall Thickness per UG16(b), $tr_{16b} = 3.5000$ mm.
 Wall Thickness, shell/head, internal pressure $tr_{b1} = 14.2022$ mm.
 Wall Thickness $tb_1 = \max(tr_{b1}, tr_{16b}) = 14.2022$ mm.
 Wall Thickness, shell/head, external pressure $tr_{b2} = 2.3389$ mm.
 Wall Thickness $tb_2 = \max(tr_{b2}, tr_{16b}) = 3.5000$ mm.
 Wall Thickness per table UG-45 $tb_3 = 10.1026$ mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[10.103, \max(14.2022, 3.5)]$
 $= 10.1026$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(4.7968, 10.1026)$
 $= 10.1026$ mm.

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	39.9,	Allowable :	138.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	305.1 N./mm ²	Passed
Occasional :	18.1,	Allowable :	183.5 N./mm ²	Passed
Shear :	16.3,	Allowable :	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 12.0$, $t_r = 2.797$, $c = 2.0$ mm., $E^* = 1.0$
 Thickness Ratio $= t_r * E^* / (t_g - c) = 0.28$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 20.0$, $t_r = 2.797$, $c = 2.0$ mm., $E^* = 1.0$
 Thickness Ratio $= t_r * E^* / (t_g - c) = 0.155$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $36.66/51.00 = 0.719$

Weld Size Calculations, Description: N58"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 18.0000$ mm.

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (25.283 - 24.445 + 2 \cdot 18.0 \cdot 1.0 \cdot (1.0 \cdot 24.0 - 12.2022)) \cdot 138)$
 $= 7155.69 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (13.6829 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 20661.12 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2(\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$
 $= (13.6829 + 0.0 + 1. + (8.64)) \cdot 138$
 $= 32818.96 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+(2 \cdot (\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$
 $= (13.6829 + 0.0 + 1. + 0.0) \cdot 138$
 $= 32818.96 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 243.2 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 26340. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 112.6) \cdot (20.0 - 2.0) \cdot 0.7 \cdot 138$
 $= 62719. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 243.2 \cdot (20.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 79559. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (26340 + 62719) = 89060 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (26340 + 0 + 79559 + 0) = 105899 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (26340 + 79559 + 0) = 105899 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 89059 Kgf, must exceed W = 7155 Kgf	or W1 = 20661 Kgf
Path 2-2 = 105899 Kgf, must exceed W = 7155 Kgf	or W2 = 32818 Kgf
Path 3-3 = 105899 Kgf, must exceed W = 7155 Kgf	or W3 = 32818 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 46 bars



Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 16.7409 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 242.7409 mm.

Input Echo, WRC297 Item 1, Description: N58" :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	2.0000	mm.	
Shell External Corrosion Allowance	0.0000	mm.	
Shell Diameter	900.000	mm.	
Shell Thickness	26.0000	mm.	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	138.00	N./mm2
Vessel Hot S.I. Allowable	Smh	138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	2.0000	mm.	
Nozzle Diameter	203.200	mm.	
Nozzle Thickness	20.0000	mm.	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-105		
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	36.66	bars
Radial Load	P	-1305.32	Kgf
Circumferential Shear	Vc	978.99	Kgf
Longitudinal Shear	Vl	1305.32	Kgf
Circumferential Moment	Mc	521.93	Kg-m.
Longitudinal Moment	Ml	678.51	Kg-m.
Torsional Moment	Mt	782.89	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 243.2 + 2 \cdot 1.65 \cdot \sqrt{(464.0(26.0 - 2.0))}$
 $= 591.440 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	928.000	mm.
Nozzle Outside Diameter	(d) =	243.200	mm.
Vessel Thickness used	(T) =	24.000	mm.
Nozzle Thickness used	(t) =	18.000	mm.
T / t	=	1.333	
d / t	=	13.511	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.630$$

Nr/P	=	0.097
Mr/P	=	0.128
M0/P	=	0.051



NØ/P = 0.126
 MrD/Mc = 0.242
 NrDL/Mc = 0.144
 MØd/Mc = 0.112
 NØDL/Mc = 0.072
 MrD/Ml = 0.144
 NrDL/Ml = 0.095
 MØD/Ml = 0.069
 NØDL/Ml = 0.165

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	67	70	67	70
Outplane Stress Summary	49	78	104	55

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-40	40	40	-40
Inplane Pressure Stress	33	33	33	33
Inplane Stress Summary	7	55	98	-17

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	49	79	105	72

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-2	-2	2	2
Outplane Bending (Mc)	-24	24	24	-24
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	33	33	33	33



Outplane Stress Summary 16 51 70 7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to
 circumferential plane)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	17	-17	17	-17
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-53	53	53	-53
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	67	70	67	70

Inplane Stress Summary 28 103 144 8

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to
 circumferential plane)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary 2 2 4 4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress 28 103 145 12

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses in the
 hoop direction)

	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	23	27	23	27

Hoop Stress Summary 18 22 33 38

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-48	48	48	-48
Axial Pressure Stress	11	11	11	11

Axial Stress Summary -27 32 90 -44

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
 (Shear stress)

	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside



Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4
Shear Stress Summary	5	5	3	3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	47	35	90	83

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-2	-2	2	2
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	23	27	23	27
Hoop Stress Summary	23	27	28	33

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	18	-18	18	-18
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-66	66	66	-66
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	11	11	11	11
Axial Stress Summary	-42	52	105	-64

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	4	4	4	4
Shear Stress Summary	2	2	6	6

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	66	52	106	98

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	67	70	67	70	67	70	67	70
Circ. Pl (SUS)	-5	-5	9	9	-3	-3	7	7
Circ. Q (SUS)	-13	13	25	-25	-36	36	70	-70



Long. Pm (SUS)	33	33	33	33	33	33	33	33
Long. Pl (SUS)	-2	-2	6	6	0	0	4	4
Long. Q (SUS)	-23	23	57	-57	-18	18	30	-30
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	67.0	70.0	67.0	70.0	67.0	70.0	67.0	70.0
Pm+Pl (SUS)	62.0	65.0	76.0	79.0	64.0	67.0	74.0	77.0
Pm+Pl+Q (Total)	49.4	78.6	101.7	72.1	28.3	103.1	144.2	11.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	70.00	138.00	Passed
Pm+Pl (SUS)	79.02	207.00	Passed
Pm+Pl+Q (TOTAL)	144.21	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	23	27	23	27	23	27	23	27
Circ. Pl (SUS)	-5	-5	9	9	0	0	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-9	-9	9	9	-7	-7	7	7
Long. Q (SUS)	-30	30	66	-66	-48	48	84	-84
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	23.0	27.0	23.0	27.0	23.0	27.0	23.0	27.0
Pm+Pl (SUS)	18.1	22.0	32.1	36.1	23.1	27.0	27.1	31.1
Pm+Pl+Q (Total)	47.1	34.1	86.2	82.2	67.3	52.4	102.3	97.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	27.00	138.00	Passed
Pm+Pl (SUS)	36.06	207.00	Passed
Pm+Pl+Q (TOTAL)	102.33	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Input, Nozzle Desc: K1A3"

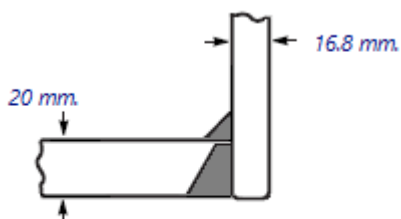
From: 40

Pressure for Reinforcement Calculations	P	36.658	bars
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	120	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	900.00	mm.
Design Length of Section	L	3150.0000	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1A3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 36.66 \cdot 452.0 / (138 \cdot 1.0 - 0.6 \cdot 36.66)$
 $= 12.2022$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 27.4 \cdot \exp([36.66 / (138 \cdot 1.0)] - 1)$
 $= 0.7377$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3924 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	120.3280	mm.
Parallel to Vessel Wall	Rn+tn+t	60.1640	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.9100	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1A3".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.7377	mm.
Wall Thickness per UG16(b),	tr16b = 3.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1 = 14.2022	mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 14.2022	mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3389	mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000	mm.
Wall Thickness per table UG-45	tb3 = 6.8000	mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.8, \max(14.2022, 3.5)]$
 $= 6.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.7377, 6.8)$
 $= 6.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 16.764, tr = 0.738, c = 2.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.05$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :



Design Pressure/Ambient Rating = $36.66/51.00 = 0.719$

Weld Size Calculations, Description: K1A3"

Intermediate Calc. for nozzle/shell Welds T_{min} 14.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.9797 mm.
The Cut Length for this Nozzle is, Drop + H_o + H + T : 228.3547 mm.

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Input, Nozzle Desc: K1B3"

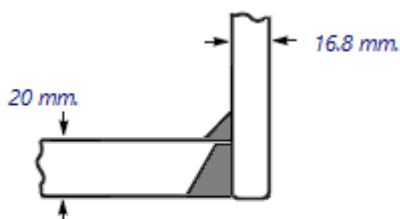
From: 40

Pressure for Reinforcement Calculations	P	36.658	bars
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	120	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	900.00	mm.
Design Length of Section	L	3150.0000	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		800.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K1B3"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 3.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 36.66 \cdot 452.0 / (138 \cdot 1.0 - 0.6 \cdot 36.66)$
 $= 12.2022$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 36.66 \cdot 40.1 / (138 \cdot 1.0 - 0.6 \cdot 36.66)$
 $= 1.0825$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.4518 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	160.4000	mm.
Parallel to Vessel Wall, opening length	d	80.2000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.9100	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

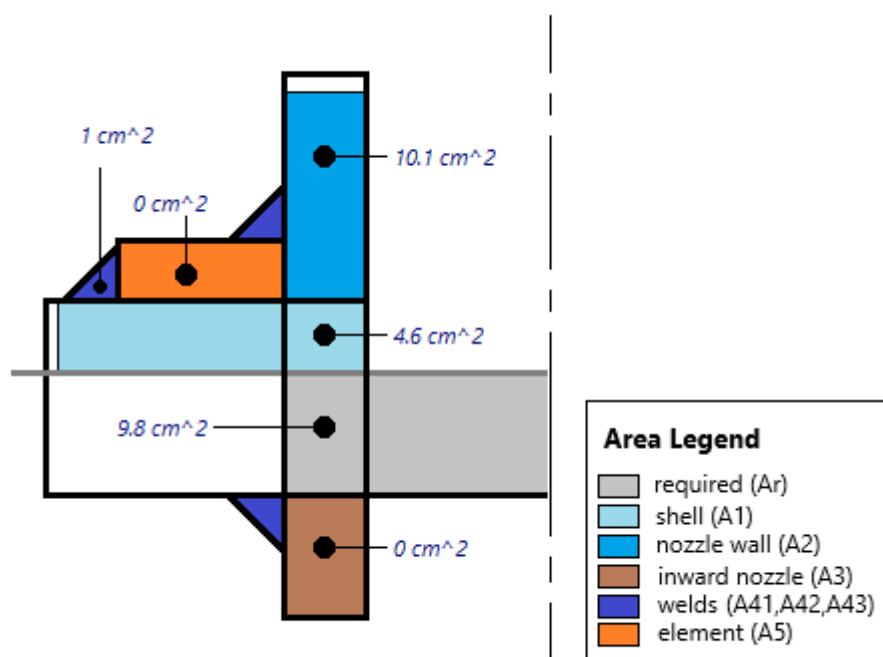


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5	MAWP	External	Mapnc
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Area Required	Ar	9.786	1.999	NA
Area in Shell	A1	4.650	10.439	NA
Area in Nozzle Wall	A2	10.100	10.565	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	15.749	22.004	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (80.2 \cdot 12.2022 \cdot 1.0 + 2 \cdot 14.764 \cdot 12.2022 \cdot 1.0 \cdot (1 - 1.0))$
 $= 9.786 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 80.2(1.0 \cdot 18.0 - 1.0 \cdot 12.202) - 2 \cdot 14.764$
 $(1.0 \cdot 18.0 - 1.0 \cdot 12.202) \cdot (1 - 1.0)$
 $= 4.650 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 36.91)(14.76 - 1.08)1.0$
 $= 10.100 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 3.0825 \text{ mm.}$
 Wall Thickness per UG16(b), $tr16b = 3.5000 \text{ mm.}$
 Wall Thickness, shell/head, internal pressure $trb1 = 14.2022 \text{ mm.}$
 Wall Thickness $tb1 = \max(trb1, tr16b) = 14.2022 \text{ mm.}$
 Wall Thickness, shell/head, external pressure $trb2 = 2.3389 \text{ mm.}$
 Wall Thickness $tb2 = \max(trb2, tr16b) = 3.5000 \text{ mm.}$
 Wall Thickness per table UG-45 $tb3 = 7.2578 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.258, \max(14.2022, 3.5)]$
 $= 7.2578 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.0825, 7.2578)$
 $= 7.2578 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.764$, $tr = 1.083$, $c = 2.0 \text{ mm.}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E*/(tg - c) = 0.073$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C
 Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit -9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 36.66/51.00 = 0.719

Weld Size Calculations, Description: K1B3"

Intermediate Calc. for nozzle/shell Welds Tmin 14.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f r_1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (9.7862 - 4.6498 + 2 \cdot 14.764 \cdot 1.0 \cdot (1.0 \cdot 18.0 - 12.2022)) \cdot 138)$
 $= 9636.68 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f r_2) \cdot S_v$
 $= (10.0997 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 15618.95 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r_1)) \cdot S_v$
 $= (10.0997 + 0.0 + 1. + (5.315)) \cdot 138$
 $= 23098.05 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$
 $= (10.0997 + 0.0 + 1. + 0.0 + (5.315)) \cdot 138$
 $= 23098.05 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 109.728 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 11884. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 47.482) \cdot (16.764 - 2.0) \cdot 0.7 \cdot 138$
 $= 21693. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 109.728 \cdot (20.0 - 2.0) \cdot 0.74 \cdot 138$
 $= 32306. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (11884 + 21693) = 33578 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$



$$= (11884 + 0 + 32306 + 0) = 44190 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (11884 + 32306 + 0) = 44190 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 33577 Kgf, must exceed W = 9636 Kgf or W1 = 15618 Kgf

Path 2-2 = 44190 Kgf, must exceed W = 9636 Kgf or W2 = 23098 Kgf

Path 3-3 = 44190 Kgf, must exceed W = 9636 Kgf or W3 = 23098 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 47 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 3.3570 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.3570 mm.

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Input, Nozzle Desc: K2A2"

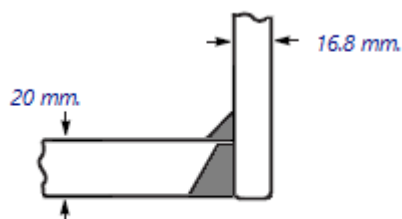
From: 40

Pressure for Reinforcement Calculations	P	36.711	bars
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	120	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	900.00	mm.
Design Length of Section	L	3150.0000	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 36.71 \cdot 452.0 / (138 \cdot 1.0 - 0.6 \cdot 36.71)$
 $= 12.2202$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 27.4 \cdot \exp([36.71 / (138 \cdot 1.0)] - 1)$
 $= 0.7387$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3871 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	120.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	60.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.9100 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2A2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.7387 mm.
Wall Thickness per UG16(b),	tr16b = 3.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 14.2202 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 14.2202 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3389 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm.
Wall Thickness per table UG-45	tb3 = 6.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.8, \max(14.2202, 3.5)]$
 $= 6.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.7387, 6.8)$
 $= 6.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 16.764, tr = 0.739, c = 2.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.05$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit	-9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 36.71/51.00 = 0.720

Weld Size Calculations, Description: K2A2"

Intermediate Calc. for nozzle/shell Welds Tmin 14.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.9797 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.9797 mm.

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Input, Nozzle Desc: K2B2"

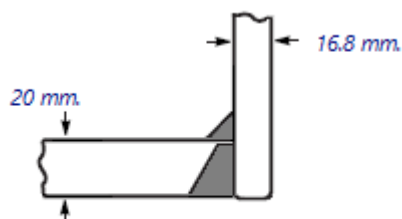
From: 40

Pressure for Reinforcement Calculations	P	36.711	bars
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	120	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	900.00	mm.
Design Length of Section	L	3150.0000	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		200.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: K2B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 36.71 \cdot 452.0 / (138 \cdot 1.0 - 0.6 \cdot 36.71)$
 $= 12.2202$ mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 27.4 \cdot \exp([36.71 / (138 \cdot 1.0)] - 1)$
 $= 0.7387$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 0.3871 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	120.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	60.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.9100 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2B2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.7387 mm.
Wall Thickness per UG16(b),	tr16b = 3.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 14.2202 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 14.2202 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3389 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm.
Wall Thickness per table UG-45	tb3 = 6.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.8, \max(14.2202, 3.5)]$
 $= 6.8000$ mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.7387, 6.8)$
 $= 6.8000$ mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 16.764, tr = 0.739, c = 2.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.05$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit	-9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 36.71/51.00 = 0.720

Weld Size Calculations, Description: K2B2"

Intermediate Calc. for nozzle/shell Welds Tmin 14.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.9797 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.9797 mm.

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Input, Nozzle Desc: M124"

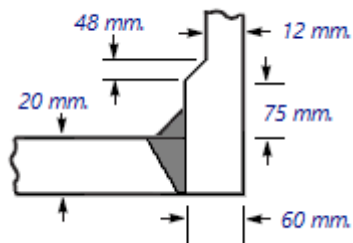
From: 40

Pressure for Reinforcement Calculations	P	36.658	bars
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	120	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	900.00	mm.
Design Length of Section	L	3150.0000	mm.
Shell Finished (Minimum) Thickness	t	20.0000	mm.
Shell Internal Corrosion Allowance	cas	2.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	48.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		60.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	133.3760	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	14.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: M124"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 24.000 in.
 Actual Thickness Used in Calculation 0.472 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 36.66 \cdot 452.0 / (138 \cdot 1.0 - 0.6 \cdot 36.66)$
 $= 12.2022 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 36.66 \cdot 306.8 / (138 \cdot 1.0 - 0.6 \cdot 36.66)$
 $= 8.2824 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0606 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 48.0 / 0.5773$
 $= 83.1385 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

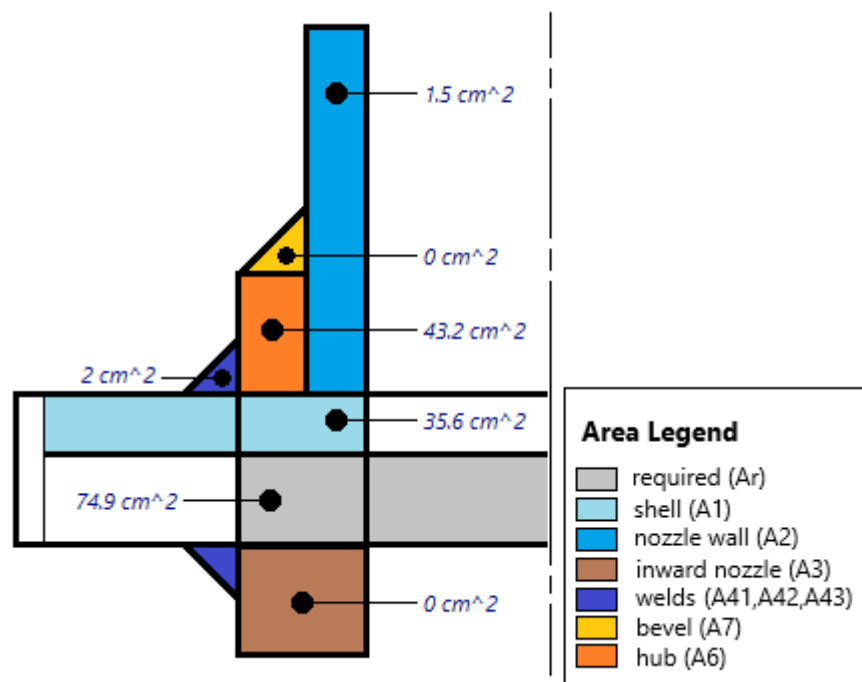
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1227.2000 mm.
Parallel to Vessel Wall, opening length	d	613.6000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	45.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	74.873	15.291	NA
Area in Shell	A1	35.575	79.867	NA
Area in Nozzle Wall	A2	1.546	8.045	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.960	1.960	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	43.200	43.200	NA
TOTAL AREA AVAILABLE	Atot	82.281	133.072	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (613.6 \cdot 12.2022 \cdot 1.0 + 2 \cdot 58.0 \cdot 12.2022 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 74.873 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 613.6(1.0 \cdot 18.0 - 1.0 \cdot 12.202) - 2 \cdot 58.0$$

$$(1.0 \cdot 18.0 - 1.0 \cdot 12.2022) \cdot (1 - 1.0)$$

$$= 35.575 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 45.0)(10.0 - 8.28)1.0$$

$$= 1.546 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 14.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.960 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(45.0, 133.4, 75.0)) \cdot (60.0 - 12.0) \cdot 1.0$$

$$= 43.200 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 10.2824 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 10.2824 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 12.0$, $t_r = 8.282$, $c = 2.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.828$, Temp. Reduction = $10 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 $^\circ\text{C}$
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-19 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-29 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, $t_g = 20.0$, $t_r = 12.202$, $c = 2.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.678$, Temp. Reduction = $18 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	12 $^\circ\text{C}$
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-4 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-22 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : -22 $^\circ\text{C}$

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 $^\circ\text{C}$
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-33 $^\circ\text{C}$

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $36.66/51.00 = 0.719$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (Hub \text{ Thickness} - Neck \text{ Thickness}) / \tan(30)$$

$$= 48.0/0.5773$$

$$= 83.1385 \text{ mm.}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	920.4000 mm.
Parallel to Vessel Wall, opening length	d	460.2000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	45.0000 mm.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	49.918	49.918
Area Available in Shell	A1	17.788	17.788
Area Available in Nozzle Wall	A2	1.546	1.546
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	1.960	1.960

Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	43.200	43.200
TOTAL AREA AVAILABLE	Atot	64.493	64.493

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: M124"

Intermediate Calc. for nozzle/shell Welds Tmin 18.0000 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 9.8980 = $0.7 \cdot W_o$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (74.8728 - 35.5752 + 2 \cdot 58.0 \cdot 1.0 \cdot (1.0 \cdot 18.0 - 12.2022)) \cdot 138)$
 $= 64761.73 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (44.7458 + 1.96 - 0.0 \cdot 1.0) \cdot 138$
 $= 65722.45 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (44.7458 + 0.0 + 1.96 + (20.88)) \cdot 138$
 $= 95103.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (44.7458 + 0.0 + 1.96 + 0.0 + (20.88)) \cdot 138$
 $= 95103.88 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 729.6 \cdot 14.0 \cdot 0.49 \cdot 138$
 $= 110630. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 335.8) \cdot (60.0 - 2.0) \cdot 0.7 \cdot 138$
 $= 602697. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{g_{nvi}} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 729.6 \cdot (14.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 167073. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (110630 + 602697) = 713327 \text{ Kgf}$

Failure Path [Path 2-2]:



$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$
$$= (110630 + 0 + 167073 + 0) = 277703 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$
$$= (110630 + 167073 + 0) = 277703 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 713326 Kg, must exceed W = 64761 Kg or W1 = 65722 Kg
Path 2-2 = 277702 Kg, must exceed W = 64761 Kg or W2 = 95103 Kg
Path 3-3 = 277702 Kg, must exceed W = 64761 Kg or W3 = 95103 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 38 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 186.5214 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 339.8974 mm.

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 512.77 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 613.60 mm.

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Input, Nozzle Desc: N216"

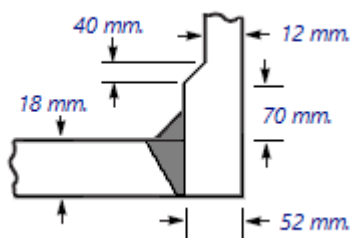
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Pressure for Reinforcement Calculations	P	36.658	bars
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	120	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	900.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	18.0000	mm.
Head Internal Corrosion Allowance	cas	2.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm.
Height of Beveled Transition	L'	40.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		52.0000	mm.
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	462.4740	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).





Reinforcement CALCULATION, Description: N216"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 16.000 in.
 Actual Thickness Used in Calculation 0.472 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 36.66 \cdot 904.0 \cdot 0.994 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 36.66)$
 $= 11.9692$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 36.66 \cdot 205.2 / (138 \cdot 1.0 - 0.6 \cdot 36.66)$
 $= 5.5396$ mm.

Required Nozzle thickness under External Pressure per UG-28 : 1.4266 mm.

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0 / 0.5773$
 $= 69.2821$ mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

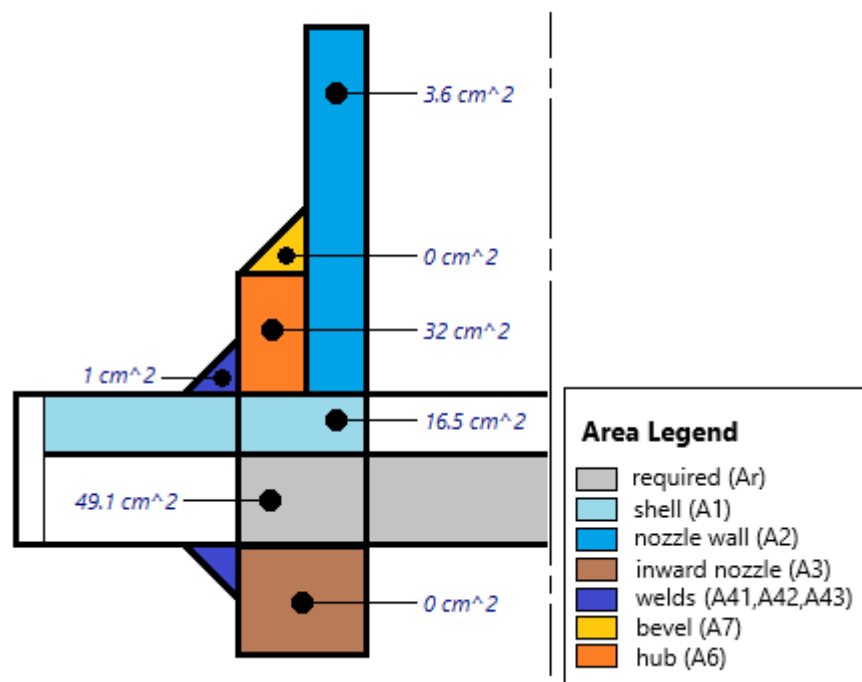
UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	820.8000	mm.
Parallel to Vessel Wall, opening length	d	410.4000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	40.0000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	49.121	4.973	NA
Area in Shell	A1	16.543	55.718	NA
Area in Nozzle Wall	A2	3.568	6.859	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	32.000	32.000	NA
TOTAL AREA AVAILABLE	Atot	53.111	95.577	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (410.4 \cdot 11.9692 \cdot 1.0 + 2 \cdot 50.0 \cdot 11.9692 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 49.121 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 410.4(1.0 \cdot 16.0 - 1.0 \cdot 11.969) - 2 \cdot 50.0$$

$$(1.0 \cdot 16.0 - 1.0 \cdot 11.9692) \cdot (1 - 1.0)$$

$$= 16.543 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 40.0)(10.0 - 5.54)1.0$$

$$= 3.568 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(40.0, 462.5, 70.0)) \cdot (52.0 - 12.0) \cdot 1.0$$

$$= 32.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 7.5396 mm.
Wall Thickness per UG16(b),	tr16b = 3.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 13.9692 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 13.9692 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3368 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm.
Wall Thickness per table UG-45	tb3 = 10.3312 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[10.331, \max(13.9692, 3.5)]$$

$$= 10.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(7.5396, 10.3312)$$

$$= 10.3312 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 59.5,	Allowable	: 138.0 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 285.5 N./mm2	Passed
Occasional	: 36.7,	Allowable	: 183.5 N./mm2	Passed
Shear	: 16.6,	Allowable	: 96.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 12.0, tr = 5.54, c = 2.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.554, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-46 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 18.0, tr = 11.969, c = 2.0 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.748, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-21 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -21 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 36.66/51.00 = 0.719

Weld Size Calculations, Description: N216"



Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (49.1214 - 16.5426 + 2 \cdot 50.0 \cdot 1.0 \cdot (1.0 \cdot 16.0 - 11.9692)) \cdot 138)$
 $= 51515.51 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (35.5683 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 51457.37 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (35.5683 + 0.0 + 1. + (16.0)) \cdot 138$
 $= 73971.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (35.5683 + 0.0 + 1. + 0.0 + (16.0)) \cdot 138$
 $= 73971.88 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 510.4 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 55280. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 230.2) \cdot (52.0 - 2.0) \cdot 0.7 \cdot 138$
 $= 356177. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 510.4 \cdot (18.0 - 2.0) \cdot 0.74 \cdot 138$
 $= 133575. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (55280 + 356177) = 411457 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (55280 + 0 + 133575 + 0) = 188855 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (55280 + 133575 + 0) = 188855 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 411457 Kgf, must exceed W = 51515 Kgf	or W1 = 51457 Kgf
Path 2-2 = 188855 Kgf, must exceed W = 51515 Kgf	or W2 = 73971 Kgf
Path 3-3 = 188855 Kgf, must exceed W = 51515 Kgf	or W3 = 73971 Kgf



Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 38 bars

Nozzle is O.K. for the External Pressure: 1 bars

The Drop for this Nozzle is : 40.4942 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 520.9681 mm.

Input Echo, WRC107/537 Item 1, Description: N216" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	2.0000	mm.
Vessel Diameter	Dv	1620.000	mm.
Vessel Thickness	Tv	18.000	mm.
Design Temperature	T1	120.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm2
Vessel Hot S.I. Allowable	Smh	138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	2.0000	mm.
Nozzle Diameter	Dn	406.400	mm.
Nozzle Thickness	Tn	52.000	mm.
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm2
Design Internal Pressure	Dp	36.658	bars
Include Pressure Thrust		No	

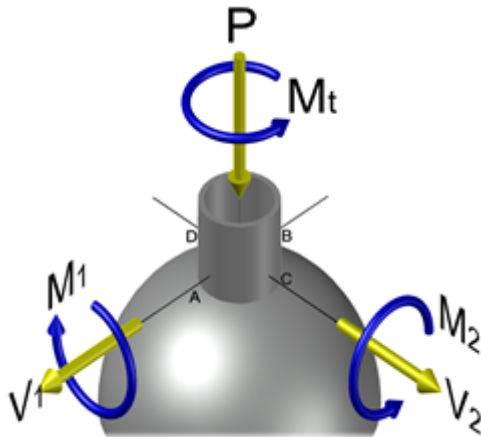
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-2610.6	Kgf
Longitudinal Shear (SUS)	(V1) V1	2307.5	Kgf
Circumferential Shear (SUS)	(Vc) V2	2307.5	Kgf
Circumferential Moment (SUS)	(Mc) M1	2421.1	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	2421.1	Kg-m.
Torsional Moment (SUS)	Mt	3131.6	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 510.4 + 2 \cdot 1.65 \cdot \sqrt{(820.0 (18.0 - 2.0))}$
 $= 888.390 \text{ mm.}$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	-2610.6	Kgf
Circumferential Shear	(VC) V2	2307.5	Kgf
Longitudinal Shear	(VL) V1	2307.5	Kgf
Circumferential Moment	(MC) M1	2421.1	Kg-m.
Longitudinal Moment	(ML) M2	2421.1	Kg-m.
Torsional Moment	MT	3131.6	Kg-m.

Dimensionless Param: U = 2.23 TAU = 5.00 (4.60) RHO = 0.32

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.01529	(A,B,C,D)
$M(x) / P$	SP 1	0.01296	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.01262	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.01342	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.01262	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.01342	(A,B,C,D)
$N(y) * T / P$	SP 1	0.02253	(A,B,C,D)
$M(y) / P$	SP 1	0.00497	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.01433	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 1	0.00406	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.01433	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 1	0.00406	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5
Rad. Bend.	P	7.8	-7.8	7.8	-7.8	7.8	-7.8	7.8	-7.8
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-10.2	-10.2	10.2	10.2
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-65.2	65.2	65.2	-65.2
Rad. Memb.	ML	-10.2	-10.2	10.2	10.2	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-65.2	65.2	65.2	-65.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-66.1	48.7	84.7	-61.2	-66.1	48.7	84.7	-61.2



Tang. Memb. P	2.3	2.3	2.3	2.3	2.3	2.3	2.3	2.3
Tang. Bend. P	3.0	-3.0	3.0	-3.0	3.0	-3.0	3.0	-3.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-11.6	-11.6	11.6	11.6
Tang. Bend. MC	0.0	0.0	0.0	0.0	-19.7	19.7	19.7	-19.7
Tang. Memb. ML	-11.6	-11.6	11.6	11.6	0.0	0.0	0.0	0.0
Tang. Bend. ML	-19.7	19.7	19.7	-19.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-26.1	7.4	36.6	-8.9	-26.1	7.4	36.6	-8.9
Shear VC	1.8	1.8	-1.8	-1.8	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Shear MT	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7
Tot. Shear	6.5	6.5	2.9	2.9	2.9	2.9	6.5	6.5
Str. Int.	67.1	49.7	84.9	61.4	66.3	48.9	85.6	62.0

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm2)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	93.0	93.0	93.0	93.0	93.0	93.0	93.0	93.0
Rad.	Pl (SUS)	-8.7	-8.7	11.7	11.7	-8.7	-8.7	11.7	11.7
Rad.	Q (SUS)	-57.4	57.4	73.0	-73.0	-57.4	57.4	73.0	-73.0
Long.	Pm (SUS)	93.0	93.0	93.0	93.0	93.0	93.0	93.0	93.0
Long.	Pl (SUS)	-9.4	-9.4	13.9	13.9	-9.4	-9.4	13.9	13.9
Long.	Q (SUS)	-16.8	16.8	22.7	-22.7	-16.8	16.8	22.7	-22.7
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.8	1.8	-1.8	-1.8	-1.8	-1.8	1.8	1.8
Shear	Q (SUS)	4.7	4.7	4.7	4.7	4.7	4.7	4.7	4.7
Pm (SUS)		93.0	93.0	93.0	93.0	93.0	93.0	93.0	93.0
Pm+Pl (SUS)		85.8	85.8	107.9	107.9	85.8	85.8	107.9	107.9
Pm+Pl+Q (Total)		67.9	142.8	177.9	84.3	67.1	142.0	178.6	85.0

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	93.05	138.00	Passed
Pm+Pl (SUS)	107.91	207.00	Passed
Pm+Pl+Q (TOTAL)	178.62	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N4_2"

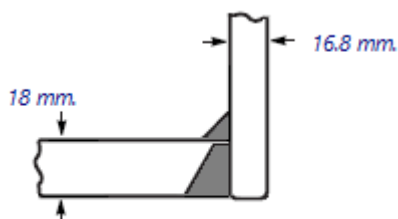
From: 50

Pressure for Reinforcement Calculations	P	36.658	bars
Temperature for Internal Pressure	Temp	120	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	120	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	900.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	18.0000	mm.
Head Internal Corrosion Allowance	cas	2.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	400.0000	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	2.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N4_2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 36.66 \cdot 904.0 \cdot 0.994 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 36.66)$
 $= 11.9692 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 27.4 \cdot (\exp([36.66 / (138.0 \cdot 1.0)] - 1))$
 $= 0.7377 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3871 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 155.1727 mm.
 Parallel to Vessel Wall, opening length d 77.5864 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 36.9100 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

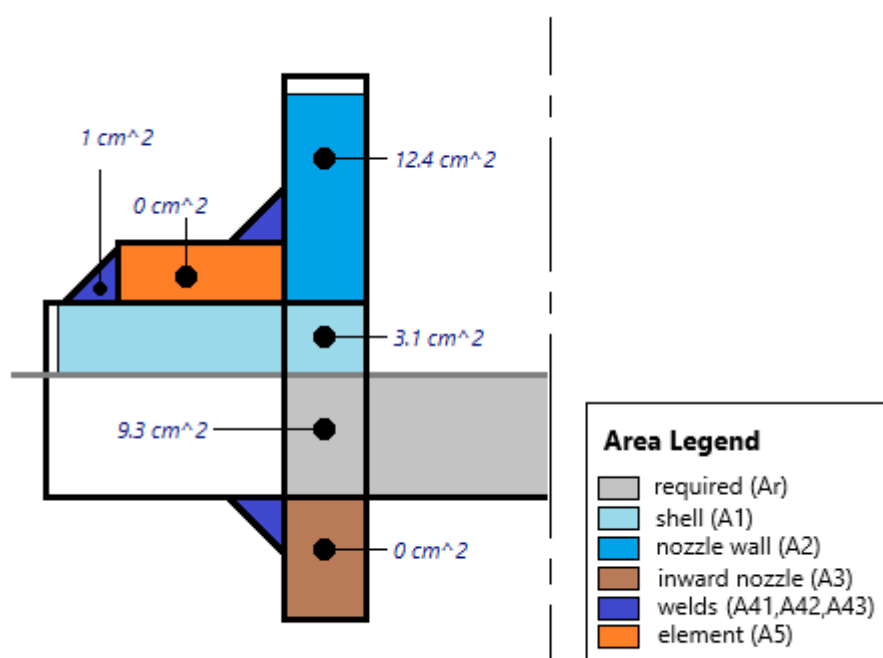


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	9.286	0.940	NA
Area in Shell	A1	3.127	10.534	NA
Area in Nozzle Wall	A2	12.394	12.704	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		1.000	1.000	NA
Area in Element	A5	0.000	0.000	NA
TOTAL AREA AVAILABLE	Atot	16.521	24.237	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 44.94 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (77.5864 \cdot 11.9692 \cdot 1.0 + 2 \cdot 14.764 \cdot 11.9692 \cdot 1.0 \cdot (1 - 1.0))$
 $= 9.286 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 77.586(1.0 \cdot 16.0 - 1.0 \cdot 11.969) - 2 \cdot 14.764$
 $(1.0 \cdot 16.0 - 1.0 \cdot 11.9692) \cdot (1 - 1.0)$
 $= 3.127 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 36.91)(14.76 - 0.74)1.0/\sin(56.7)$
 $= 12.394 \text{ cm}^2$

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000 \text{ cm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, $tg = 16.764$, $tr = 0.738$, $c = 2.0 \text{ mm.}$, $E* = 1.0$
 Thickness Ratio $= tr \cdot E*/(tg - c) = 0.05$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 7 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $36.66/51.00 = 0.719$

Weld Size Calculations, Description: N4 2"

Intermediate Calc. for nozzle/shell Welds Tmin 14.7640 mm.

Results Per UW-16.1:

Required Thickness Actual Thickness



Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (9.2864 - 3.1274 + 2 \cdot 14.764 \cdot 1.0 \cdot (1.0 \cdot 16.0 - 11.9692)) \cdot 138)$
 $= 10341.58 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (12.394 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 18847.45 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (12.394 + 0.0 + 1. + (4.7245)) \cdot 138$
 $= 25495.53 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (12.394 + 0.0 + 1. + 0.0 + (4.7245)) \cdot 138$
 $= 25495.53 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 119.3924 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 12931. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 49.2447) \cdot (16.764 - 2.0) \cdot 0.7 \cdot 138$
 $= 22499. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 119.3924 \cdot (18.0 - 2.0) \cdot 0.74 \cdot 138$
 $= 31246. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SONW + SNW) = (12931 + 22499) = 35430 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (12931 + 0 + 31246 + 0) = 44177 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (Sonw + Tngw + Sinw)$
 $= (12931 + 31246 + 0) = 44177 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 35429	Kgf, must exceed W = 10341 Kgf	or W1 = 18847 Kgf
Path 2-2 = 44176	Kgf, must exceed W = 10341 Kgf	or W2 = 25495 Kgf
Path 3-3 = 44176	Kgf, must exceed W = 10341 Kgf	or W3 = 25495 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 49 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 1 bars



Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N4_2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000 in.
Actual Thickness Used in Calculation	0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 36.66 \cdot 904.0 \cdot 0.994 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 36.66)$
 $= 11.9692 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 27.4 \cdot \exp([36.66 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7377 \text{ mm.}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3871 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	116.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	58.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	36.9100 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N4_2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.7377 mm.
Wall Thickness per UG16(b),	tr16b = 3.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 13.9692 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 13.9692 mm.
Wall Thickness, shell/head, external pressure	trb2 = 2.3368 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm.
Wall Thickness per table UG-45	tb3 = 6.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.8, \max(13.9692, 3.5)]$
 $= 6.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(2.7377, 6.8)$
 $= 6.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: N4_2"

Intermediate Calc. for nozzle/shell Welds	Tmin	14.7640 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Nozzle is O.K. for the External Pressure: 1 bars

PV Elite 28 U1 Licensee: SL User
FileName : A140NHT11GQ001_2-HP161_H2 Stroage 1st stage KO d Page 122 of 123
Nozzle Calcs.: N4__2" Nozl: 30 2:03pm Sep 17,2026



HEXAGON

The Drop for this Nozzle is : 25.0655 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 245.5836 mm.

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: One or more nozzles do not meet proper Spacing Requirements!**

Please review all reports carefully!

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??2=2



PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: